



Project Report On

**ImageGuard - Detection of Fake Images in Online
Marketplaces Using Deep Learning and Explainable
AI**

*Submitted in partial fulfillment of the requirements for the
award of the degree of*

Bachelor of Technology

in

Computer Science and Engineering

By

Aleena P (U2203026)

Alen Biju Nilakkappillil (U2203027)

Anand Basil (U2203039)

Anna Praveen (U2203050)

Under the guidance of

Dr. Preetha K. G.

**Department of Computer Science and Engineering
Rajagiri School of Engineering & Technology (Autonomous)
(Parent University: APJ Abdul Kalam Technological University)**

Rajagiri Valley, Kakkanad, Kochi, 682039

March 2026

CERTIFICATE

*This is to certify that the Project report entitled "**ImageGuard - Detection of Fake Images in Online Marketplaces Using Deep Learning and Explainable AI**" is a bonafide record of the work done by **Aleena P (U2203026)**, **Alen Biju Nilakkappillil (U2203027)**, **Anand Basil (U2203039)**, **Anna Praveen (U2203050)** submitted to the Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in "Computer Science and Engineering" during the academic year 2025-2026.*

Project Guide

Dr. Preetha K. G.

Professor & HOD

Dept. of CSE

RSET

Project Coordinator

Dr. Jisha G

Associate Professor

Dept. of CSE

RSET

Head of the Department

Dr. Preetha K. G.

Professor

Dept. of CSE

RSET

ACKNOWLEDGMENT

We wish to express our sincere gratitude towards **Rev. Dr. Jaison Paul Mulerrickal CMI**, Principal of RSET, and **Dr. Preetha K. G.**, Head of the Department of Computer Science and Engineering for providing us with the opportunity to undertake our project, "ImageGuard - Detection of Fake Images in Online Marketplaces Using Deep Learning and Explainable AI".

We are highly indebted to our project coordinator, **Dr. Jisha G**, Associate Professor, Department of Computer Science and Engineering for her valuable support.

It is indeed our pleasure and a moment of satisfaction for us to express our sincere gratitude to our project guide **Dr. Preetha K. G.**, Professor and Head of the Department of Computer Science and Engineering for her patience and all the priceless advice and wisdom she has shared with us.

Last but not least, we would like to express our sincere gratitude towards all other teachers and friends for their continuous support and constructive ideas.

Aleena P

Alen Biju Nilakkappillil

Anand Basil

Anna Praveen

Abstract

In recent years, online marketplaces have seen a sharp increase in the use of fake, edited, and AI-generated product images. This has led to widespread consumer deception and a decline in brand trust. Manipulated visuals often misrepresent product quality and features, misleading customers into making uninformed purchases. To address this growing issue, ImageGuard is proposed. It is an intelligent and transparent system designed to automatically detect, classify, and explain manipulated or synthetic images on e-commerce platforms.

The proposed system uses Deep Learning (DL) and Explainable AI (XAI) techniques to achieve high accuracy and clear decision-making. We gather a carefully selected dataset of real, fake, and AI-generated fashion product images by using web scraping and manual labeling.

For classification, the EfficientNet and Vision Transformer (ViT) models capture both local and global image features. A YOLO-Regex module identifies mismatches by verifying text-image consistency. Grad-CAM and SHAP offer explainability with visual heatmaps that highlight altered areas, helping users understand the model's decisions. Finally, the trained model is launched as a browser extension. This enables real-time image verification as users browse online shopping sites. By combining advanced computer vision, natural language processing, and XAI techniques, ImageGuard aims to create a reliable, clear, and scalable system. This strengthens consumer confidence, protects brand reputation, and promotes authenticity across digital marketplaces.

Contents

Acknowledgment	i
Abstract	ii
List of Abbreviations	vi
List of Figures	viii
List of Tables	ix
1 Introduction	1
1.1 Background	1
1.2 Problem Definition	1
1.3 Scope and Motivation	1
1.4 Project Objective	2
1.5 Risks and Challenges	2
1.6 Assumptions	3
1.7 Societal/Industrial Relevance	3
1.8 Organization of the Report	4
1.9 Summary of the Chapter	5
2 Literature Survey	7
2.1 Introduction	7
2.2 Literature Review	7
2.2.1 Detecting Fake Images on Social Media using Machine Learning [1][2][3]	7
2.2.2 Deepfake Image Detection using Vision Transformer Models[4]	9
2.2.3 Harnessing eXplainable Artificial Intelligence for Feature Selection: A Comparative Analysis of Grad-CAM and SHAP [5][6][7]	10

2.2.4	Object Detection Based on YOLO Network[8][9]	12
2.2.5	Fake Image Detection Using Deep Learning[10]	12
2.3	Comparison of Literature Papers	14
2.4	Gaps Identified	15
2.5	Summary of the Chapter	15
3	System Design	16
3.1	Overall System Architecture	16
3.2	Design Diagrams	17
3.3	Component Design	18
3.4	Algorithms Used and Design	19
3.5	Software and Hardware Requirements	19
3.6	Dataset Identified	20
3.7	Work Breakdown and Responsibilities	21
3.8	Project Timeline	22
3.9	Summary	22
4	System Implementation	24
4.1	Dataset Engineering and Preprocessing	24
4.2	Deep Learning Model Development	24
4.3	Multimodal Verification Layer	26
4.4	Explainable AI (XAI) Integration	27
4.5	Real-Time Deployment Strategy	28
5	Results and Discussion	30
5.1	Experimental Results and Performance Evaluation	30
5.1.1	Performance Metrics of Deep learning Models and Model Comparison	30
5.1.2	Receiver Operating Characteristic (ROC) Curve Analysis	31
5.1.3	Training and Validation Performance	32
5.1.4	Confusion Matrix Insights	33
5.2	Explainable AI (XAI) Outcomes	33
5.2.1	Visual Analysis (Grad-CAM)	33
5.2.2	Feature Contribution (SHAP)	34

5.3	Browser Extension and Real-Time Deployment Results	35
5.3.1	Deployment and Integration Performance	35
5.3.2	Real-World Detection Outcomes	35
5.4	Multimodal Verification Results	38
5.5	Discussion	39
6	Conclusion and Future Scope	40
	References	41
	List of Publications	43
	Appendix A: Presentation	44
	Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes	79
	Appendix C: CO-PO-PSO Mapping	84

List of Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
CNN	Convolutional Neural Network
FC	Fully Connected (layer in a neural network)
GAN	Generative Adversarial Network
GPU	Graphics Processing Unit
NLP	Natural Language Processing
RAM	Random Access Memory
ReLU	Rectified Linear Unit
SHAP	SHapley Additive exPlanations
ViT	Vision Transformer
VS Code	Visual Studio Code
XAI	Explainable AI
YOLO	You Only Look Once

List of Figures

2.1	CNN Architecture Diagram	8
2.2	ViT Architecture Diagram	9
2.3	Grad-CAM Architecture Diagram	10
2.4	SHAP Architecture Diagram	11
3.1	Architecture Diagram	17
3.2	Sequence Diagram	18
3.3	Project Timeline	22
4.1	EfficientNet-B2 Architecture Diagram	25
4.2	ViT Architecture Diagram	26
4.3	Product Description Analysis Architecture Diagram	27
4.4	Browser Extension Workflow	29
5.1	EfficientNet ROC Curve	31
5.2	ViT ROC Curve	31
5.3	Training and Validation loss and Accuracy of EfficientNet	32
5.4	Training and Validation loss and Accuracy of ViT	32
5.5	EfficientNet Confusion Matrix	33
5.6	ViT Confusion Matrix	33
5.7	GradCAM+EfficientNet	34
5.8	GradCAM+ViT	34
5.9	SHAP+EfficientNet	34
5.10	SHAP+ViT	34
5.11	Detection of Fake image	36
5.12	XAI explanation for fake image	36
5.13	GradCAM+EfficientNet-Fake	36
5.14	GradCAM+ViT-Fake	36

5.15 SHAP+EfficientNet-Fake	37
5.16 SHAP+ViT-Fake	37
5.17 Detection of Real image	37
5.18 XAI explanation for real image	37
5.19 GradCAM+EfficientNet-Real	38
5.20 GradCAM+ViT-Real	38
5.21 SHAP+EfficientNet-Real	38
5.22 SHAP+ViT-Real	38
5.23 Product Description extracting	39
5.24 Product Description Analysis	39

List of Tables

2.1	Comparison of Research Papers	14
5.1	Quantitative Performance Comparison of Deep Learning Models	30

Chapter 1

Introduction

1.1 Background

Today, owing to the increasing trend of E-commerce, millions of product images are being uploaded every day, among which a major portion is edited, AI-generated, or even false information. The conventional methods of moderation are not able to identify these changes, which eventually leads to deception of customers, resulting in a loss of trust. ImageGuard is an AI-powered solution that utilizes deep learning technology along with Explainable AI (XAI) to identify false product images. ImageGuard utilizes CNNs, Vision Transformers, YOLO object detection, and Regex text analysis to identify discrepancies between product images and descriptions. The Grad-CAM algorithm and SHAP are also used for better transparency of the process. ImageGuard is a strong solution that utilizes multiple forms of analysis for better outcomes.

1.2 Problem Definition

To design and develop a deep learning-based model for detecting fake product images in online marketplaces. The system will analyze visual patterns, metadata inconsistencies, and image manipulations to differentiate between authentic and fake/AI-generated images.

1.3 Scope and Motivation

ImageGuard aims to develop an intelligent system that detects and classifies fake, edited, or AI-generated product images in online marketplaces. By combining Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs), the system examines both fine details and the overall image structure. This helps it identify visual manipulations.

It integrates Explainable AI tools like Grad-CAM and SHAP to provide transparent reasoning behind each detection.

Designed as both a browser extension for consumers and an API for e-commerce platforms, ImageGuard addresses the growing misuse of deceptive visuals that mislead buyers and erode trust. Unlike existing systems that rely solely on reviews or text, ImageGuard fills the gap with reliable, multimodal verification, combining core image manipulation detection with checks for text-image consistency. In the long term, the system can also be adapted for use in verifying visuals in news, ads, and social media, promoting a safer digital environment.

1.4 Project Objective

- **Detect Fake Images in Real Time:** Build a browser extension that uses Deep Learning (CNNs and Vision Transformers) to detect fake, tampered, or AI-generated product images on e-commerce websites in real time.
- **Provide Explainable Results:** Use Explainable AI methods like Grad-CAM and SHAP to highlight suspicious regions in the image and help users understand why an image was flagged.
- **Check Metadata Consistency:** Identify mismatches between product images and their descriptions using NLP (Regex) and object detection (YOLO).
- **Ensure Usability and Performance:** Develop a lightweight, user-friendly JavaScript-based browser extension that works efficiently with various image types, including low-quality or compressed images.

1.5 Risks and Challenges

- **Evolving Threats:** As generative AI rapidly improves, new types of fake images may become harder for the system to detect. Attackers may also try to fool the system using adversarial techniques.
- **Data & Accuracy Limitations:** The training dataset may not cover all manipulation types, which could lead to false positives (flagging real images) or false

negatives (missing fakes).

- **Performance & Maintenance:** Real-time detection might affect browsing speed, and the system will need regular updates to maintain accuracy and compatibility.
- **Privacy & Cost Concerns:** Analyzing images from users' browsers raises privacy issues, and the need for GPU support may make large-scale deployment expensive.

1.6 Assumptions

- **Model Accuracy:** The Deep Learning models (CNNs and Vision Transformers) are assumed to be effective at detecting fake, edited, or AI-generated images.
- **Reliable Explanations:** The Explainable AI tools (Grad-CAM and SHAP) will correctly highlight the suspicious parts of an image.
- **Real-Time Performance:** The system will process and display results quickly, without affecting the user's browsing experience.
- **User Adoption:** Users will install the extension and may contribute by reporting incorrect results.
- **Cross-Platform Support:** The browser extension will work well across various e-commerce websites and browsers.

1.7 Societal/Industrial Relevance

- **Trust and Transparency:** Establishes trust in the digital marketplace using Deep Learning and Explainable AI (XAI).
- **Societal Impact:**
 - Protects consumers from fake, edited, or AI-generated product images.
 - Fosters informed shopping and combats customer deception.
- **Industrial Impact:**
 - Provides scalable fraud detection and image verification for e-commerce platforms.

- Replaces inefficient manual moderation with automated AI solutions.
- Enhances platform authenticity and supports ethical selling practices.
- Safeguards brand reputation and promotes responsible AI adoption.
- Improves operational efficiency in the e-commerce sector.

1.8 Organization of the Report

This report is structured into four main chapters, each describing a distinct phase of the project **ImageGuard – Detection of Fake Images in Online Marketplaces Using Deep Learning and Explainable AI**. The chapters are organized as follows:

- **Chapter 1 – Introduction:**

This chapter gives an overview of the project. It discusses the background, problem definition, scope, objectives, assumptions, and challenges. It also highlights the importance of detecting fake images on e-commerce platforms for society and industry. Finally, it explains how the report is organized.

- **Chapter 2 – Literature Survey:**

This chapter offers a detailed look at current research papers and related works in fake image detection, object detection, and explainable AI. It compares different methods, including CNNs, Vision Transformers (ViT), YOLO, BERT, Grad-CAM, and SHAP. The chapter points out research gaps that the proposed project aims to address.

- **Chapter 3 – System Design:**

This chapter details the proposed system’s architecture and implementation strategy for detecting fake images in online marketplaces. It outlines the overall system design, including architectural and sequence diagrams, and describes the core components such as the hybrid CNN-ViT model, the YOLO-Regex verification module, and the Grad-CAM & SHAP explainability framework. The chapter also specifies the software and hardware requirements, the identified dataset, module responsibilities, expected outputs, and the project timeline, providing a complete technical blueprint for the project.

- **Chapter 4 – System Implementation:**

This chapter details the technical realization of the ImageGuard framework, focusing on the end-to-end pipeline from data engineering to real-time deployment. It covers dataset engineering and preprocessing, the development of the EfficientNet CNN and Vision Transformer models, the multimodal verification layer, and the integration of the explainable AI modules. Finally, it describes the deployment strategy using a browser extension and FastAPI backend.

- **Chapter 5 – Results and Discussion:**

This chapter provides a rigorous assessment of the ImageGuard framework using standard classification and diagnostic metrics. It provides a performance analysis of the CNN and ViT models, including accuracy, ROC AUC scores, and confusion matrix insights. The chapter also discusses the qualitative outcomes of the Grad-CAM and SHAP explainability analysis, along with the results of the multimodal verification for e-commerce listings.

- **Chapter 4 – Conclusion:**

The final chapter summarizes the key findings and achievements of the current phase. It also outlines the plan for the next semester, focusing on implementation, testing, and performance evaluation, along with suggestions for future enhancements.

The report concludes with references to relevant research works and appendices that include supporting materials such as presentations, programme outcomes, and CO-PO-PSO mappings.

1.9 Summary of the Chapter

Online marketplaces often show fake or altered product images. This can mislead customers and lower trust in online shopping. Current systems do not have effective tools to detect AI-generated or altered visuals.

ImageGuard is created to address this issue using Deep Learning and Explainable AI (XAI). It uses CNNs and Vision Transformers (ViT) to spot fake or modified images. It also employs Grad-CAM and SHAP to explain visually why an image was flagged. The

system includes YOLO and Regex models to check that product images match their text descriptions.

ImageGuard can work as a browser extension or API. This allows for real-time detection of fake images and helps build transparency and trust in e-commerce platforms.

Chapter 2

Literature Survey

2.1 Introduction

With the rapid growth of online marketplaces, the authenticity of product images is a major concern. Fake, manipulated, or AI-generated visuals mislead consumers and hurt the credibility of e-commerce platforms. Traditional manual checks are not enough due to the scale and complexity of modern image manipulation.

Recent advances in Deep Learning (DL) and Explainable Artificial Intelligence (XAI) have made it possible to automatically detect fake images with clear results. Models like Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) effectively handle image forensics. Explainable AI tools such as Grad-CAM and SHAP make models more transparent. Moreover, object detection models like YOLO and multimodal analysis using BERT improve the understanding of image-text relationships.

This chapter reviews important research papers that form the basis of the ImageGuard system. It compares their methods, points out research gaps, and summarizes the insights guiding this project.

2.2 Literature Review

2.2.1 Detecting Fake Images on Social Media using Machine Learning [1][2][3]

This research looks at how Convolutional Neural Networks (CNNs) can detect fake or manipulated images on social media. The authors developed a CNN-based model that can tell the difference between real and fake images by learning important features on its own. The CNN structure processes images through several layers, including convolutional, activation, and pooling layers. This setup helps the model capture low-level features like edges, color patterns, and textures, as well as high-level semantic structures such as

shapes, objects, and contextual patterns.

The study highlights how deep learning can be more effective than traditional feature extraction methods like SIFT or HOG. Instead of relying on manually crafted features, the model learns meaningful patterns from large datasets automatically. The team assessed the model's performance on different datasets. They saw a significant improvement in classification accuracy and strength against image noise and compression artifacts.

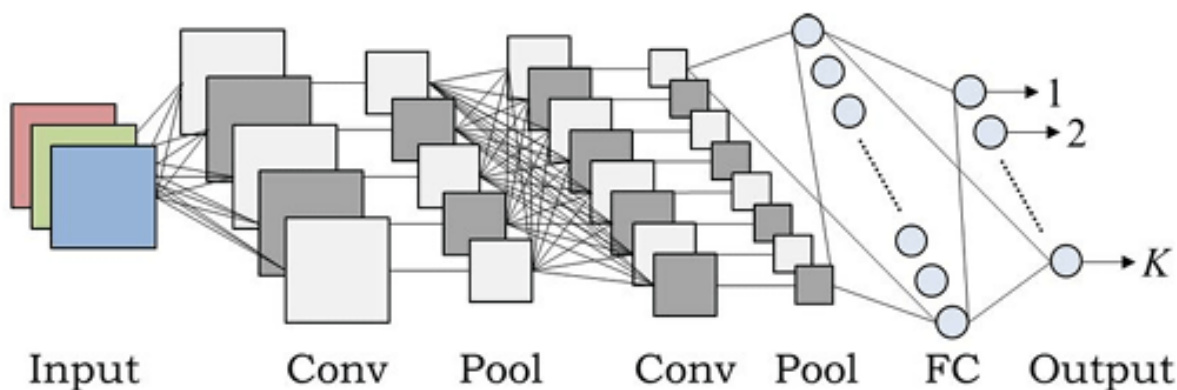


Figure 2.1: CNN Architecture Diagram

Advantages:

- Demonstrates high accuracy in complex classification tasks involving visual forensics.
- Eliminates the need for manual feature design, reducing preprocessing overhead.
- Performs well under variations in lighting, orientation, and color distortions.

Limitations:

- Requires large-scale, labeled datasets for effective training and generalization.
- Computationally expensive due to high GPU memory requirements and long training times.

Relevance to Project: This paper lays a solid groundwork for the ImageGuard project. It shows that CNN architectures can effectively detect changes in digital images. The results support using CNN models such as EfficientNet for spotting pixel-level tampering within ImageGuard. The emphasis is on finding localized inconsistencies and visual issues in product images across e-commerce platforms.

2.2.2 Deepfake Image Detection using Vision Transformer Models[4]

This study introduces Vision Transformers (ViTs) for detecting deepfake images. Unlike CNNs, which depend on local receptive fields and convolution operations, ViTs handle images by breaking them into fixed-size patches. These patches are embedded into vectors. The model then uses multi-head self-attention mechanisms to learn global relationships between various parts of the image. This approach enables the model to grasp not only local textures but also global context, which is vital for spotting high-quality manipulations like deepfakes.

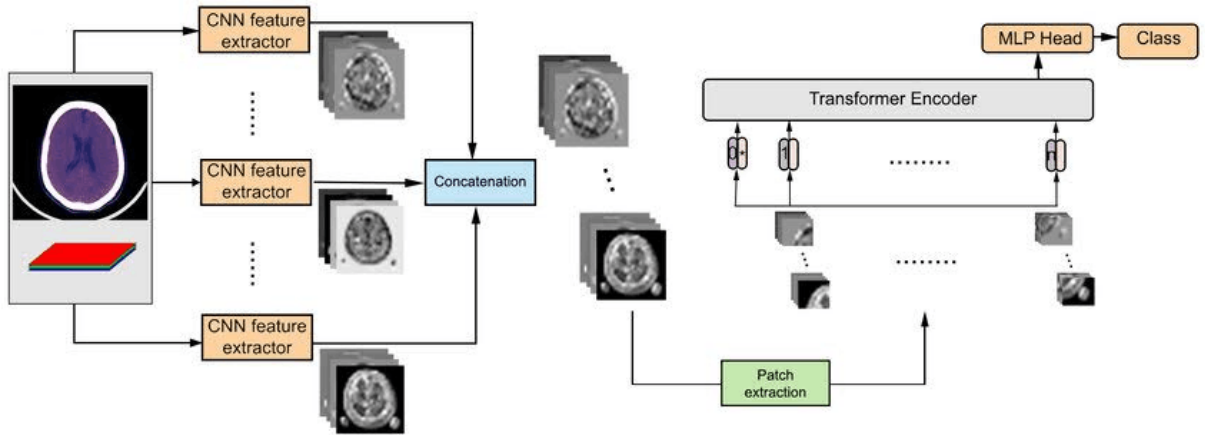


Figure 2.2: ViT Architecture Diagram

Advantages:

- Efficiently captures global image semantics and context across entire images.
- Demonstrates superior generalization when trained on large, diverse datasets.
- Performs well in detecting deepfakes that require contextual understanding beyond local patterns.

Limitations:

- High computational and memory requirements during both training and inference.
- Requires a large amount of labeled training data for optimal performance.

Relevance to Project: This research supports adding Vision Transformers to ImageGuard to work alongside CNNs. CNNs concentrate on pixel-level changes. In contrast,

ViTs help by identifying globally inconsistent features, such as mismatched lighting or unrealistic object relationships. This allows ImageGuard to manage both localized and overall image analysis effectively.

2.2.3 Harnessing eXplainable Artificial Intelligence for Feature Selection: A Comparative Analysis of Grad-CAM and SHAP [5][6][7]

This paper compares two leading Explainable Artificial Intelligence (XAI) methods: Gradient-weighted Class Activation Mapping (Grad-CAM) and SHapley Additive exPlanations (SHAP). The study shows how these techniques improve the understanding of machine learning models by explaining the reasoning behind model predictions.

Grad-CAM uses gradients from the last convolutional layers of a CNN to create a heatmap. This heatmap visually highlights the most important parts of the image that influenced the model’s decision.

SHAP relies on game theory for its analysis. It treats each image region as a ”player” and calculates how much each one contributes to the final prediction. Regions with higher scores have a greater impact on the outcome.

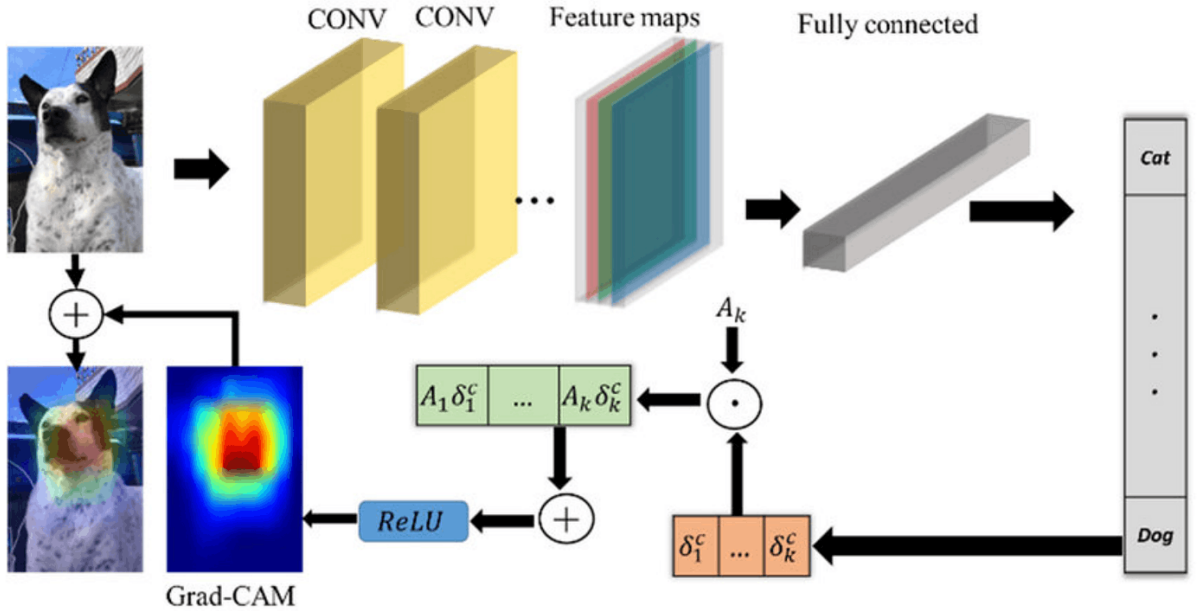


Figure 2.3: Grad-CAM Architecture Diagram

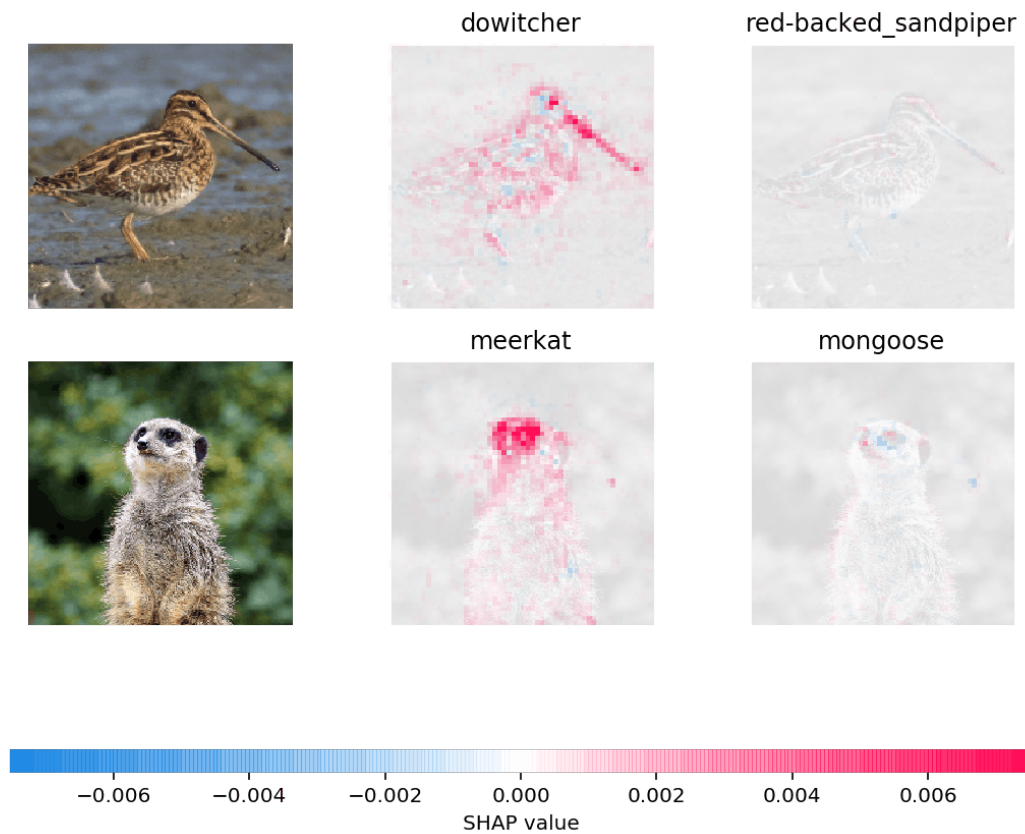


Figure 2.4: SHAP Architecture Diagram

Advantages:

- Provides intuitive visual and numerical explanations for model outputs.
- Enhances trust and interpretability of deep learning systems.
- Helps identify biases and weaknesses in trained models.

Limitations:

- Introduces additional computational overhead and latency.
- Interpretation quality depends on the complexity of the model and input data.

Relevance to Project: The combination of Grad-CAM and SHAP in ImageGuard improves explainability and user confidence. These tools will generate real-time visual explanations, or heatmaps, that show which parts of an image are flagged as suspicious. This helps users and moderators understand why an image is identified as fake. This clarity is essential for gaining user trust and promoting the use of an automated fake detection tool.

2.2.4 Object Detection Based on YOLO Network[8][9]

This paper gives an overview of the YOLO (You Only Look Once) framework, which is a real-time object detection system. It combines object localization and classification in one pass through a neural network. Unlike traditional multi-stage detectors like R-CNN or Fast R-CNN, YOLO divides the input image into a grid. Each cell in the grid predicts bounding boxes and class probabilities at the same time. **Advantages:**

- Achieves real-time object detection with remarkable speed and accuracy.
- Simple and end-to-end trainable architecture.
- Performs well across diverse image domains, including low-quality visuals.

Limitations:

- Less effective for small, partially occluded, or overlapping objects.
- Sensitive to image distortions like rotation or cropping.

Relevance to Project: YOLO is important for ImageGuard. It helps with checking objects and looking at context. It makes sure product images match their descriptions by spotting expected items. For example, if a product title says “red shoes” but YOLO finds “handbags,” the system can mark the image as suspicious. This boosts the quality of e-commerce listings and aids in automated moderation.

2.2.5 Fake Image Detection Using Deep Learning[10]

Fake Image Detection using Deep Learning refers to the use of neural network models like Convolutional Neural Networks (CNNs) and Transformers to automatically identify manipulated or AI-generated images. These systems analyze visual patterns, textures, edges, lighting inconsistencies, and structural issues that can be hard for people to notice. By training on large datasets with both real and altered images, deep learning models learn to spot subtle artifacts introduced during editing, image creation, or generative AI processes. This technology is increasingly used in areas like social media monitoring, digital forensics, online marketplaces, and security systems to prevent misinformation and fraud.

Advantages:

- Deep learning models can detect subtle visual inconsistencies that are difficult for humans to identify.
- These systems can analyze large volumes of images quickly without manual inspection.
- Models can be retrained with new datasets to detect emerging types of AI-generated or manipulated images.

Limitations:

- Deep learning models require large labeled datasets for effective training.
- Training and deploying deep learning models often require powerful GPUs and significant computational resources.

Relevance to Project: Fake image detection plays an important role in maintaining trust in digital content. It helps prevent misinformation on social media, protects consumers from fraud in online marketplaces, and supports digital forensics in identifying manipulated media. As AI-generated content becomes more widespread, reliable detection systems are essential for ensuring transparency, authenticity, and security in the digital ecosystem.

2.3 Comparison of Literature Papers

Table 2.1: Comparison of Research Papers

Paper Title	Methodology	Strengths	Limitations
Detecting Fake Images on Social Media using Machine Learning [1]	Extracts features using convolutional layers, pooling, and activation functions.	Learns visual patterns automatically; high accuracy in image classification.	Needs large datasets and GPU support.
Deepfake Image Detection using Vision Transformer Models [2]	Patches the image, encodes the position of each patch, and applies multi-head attention and MLP layers.	Understands full image context accurately; better generalization.	Requires more data; high computational cost.
Harnessing eXplainable Artificial Intelligence for Feature Selection: A Comparative Analysis of Grad-CAM and SHAP [3]	Analyzes data by making model predictions transparent and comprehensible.	Provides clear visual explanations for model decisions.	Adds extra steps and complexity to the system.
An Introduction to Image Classification and Object Detection using YOLO Detector [4]	YOLO divides images into a grid, predicting bounding boxes and class labels.	Fast and accurate object detection in one step.	May miss small or overlapping objects.
Fake Image Detection Using Deep Learning [5]	EfficientNet uses a compound scaling method to uniformly scale depth, width, and input resolution to achieve better performance with optimal efficiency.	Provides high classification accuracy with fewer parameters and lower computational cost	Requires careful hyperparameter tuning and higher computational resources for larger variants like B4–B7

2.4 Gaps Identified

The literature review shows that there are strong, specialized tools for image classification (CNNs, ViT), object detection (YOLO), and model explainability (XAI). However, there is no integrated system that combines these abilities. The main issue is the lack of a unified solution aimed at tackling fake product images in e-commerce. This solution would perform multimodal analysis (image and text) and offer user-friendly, explainable results. The ImageGuard project aims to address this gap by creating a single, easy-to-use tool that handles all these functions.

2.5 Summary of the Chapter

This chapter reviewed the key technologies that shape the design of the ImageGuard system. We looked at the literature on CNNs, Vision Transformers, XAI methods, YOLO, to outline their strengths and weaknesses. The analysis revealed a notable gap: the need for a combined, explainable, multimodal system designed to detect misleading product images in online marketplaces. The next chapter will explain the system design proposed to fill this gap.

Chapter 3

System Design

3.1 Overall System Architecture

The overall system design of ImageGuard combines several deep learning and explainable AI parts to identify fake or altered product images in online marketplaces.

The system architecture of ImageGuard is built like a pipeline. It begins with data collection and ends with a browser extension that users can access. The overall flow is as follows:

1. Data Input: A dataset of real, fake, and AI-generated product images is collected, along with their text descriptions.
2. Preprocessing: Both images and text go through preprocessing. Images are resized, normalized, and improved. Text descriptions are cleaned, broken down into tokens, and embedded.
3. Deep Learning Model: The preprocessed data goes into a deep learning model made up of an EfficientNet (CNN) and a Vision Transformer (ViT). This model classifies the image as real or fake.
4. Explainable AI (XAI): If an image is flagged as suspicious, it goes to the XAI module. Grad-CAM and SHAP generate heatmaps that highlight the areas affecting the model's decision.
5. Product Description verification: Analyzes the product description using regex-based NLP techniques and compares extracted attributes with visual outputs from YOLO object detection and color analysis. A weighted scoring algorithm calculates the match score and risk score to determine whether the product listing is legitimate or potentially fraudulent.

6. Browser Extension: The final output, which includes a verdict, confidence score, and explanatory heatmap, is shown to the user through a browser extension in real-time.

3.2 Design Diagrams

The results are shown to the user in real time. They indicate whether the image is real, fake, or does not match its description.

- Architecture Diagram – This diagram shows the high-level flow of data through the system. It starts with the initial dataset, then moves to preprocessing, the deep learning model, the XAI module, and finally ends with the browser extension output.

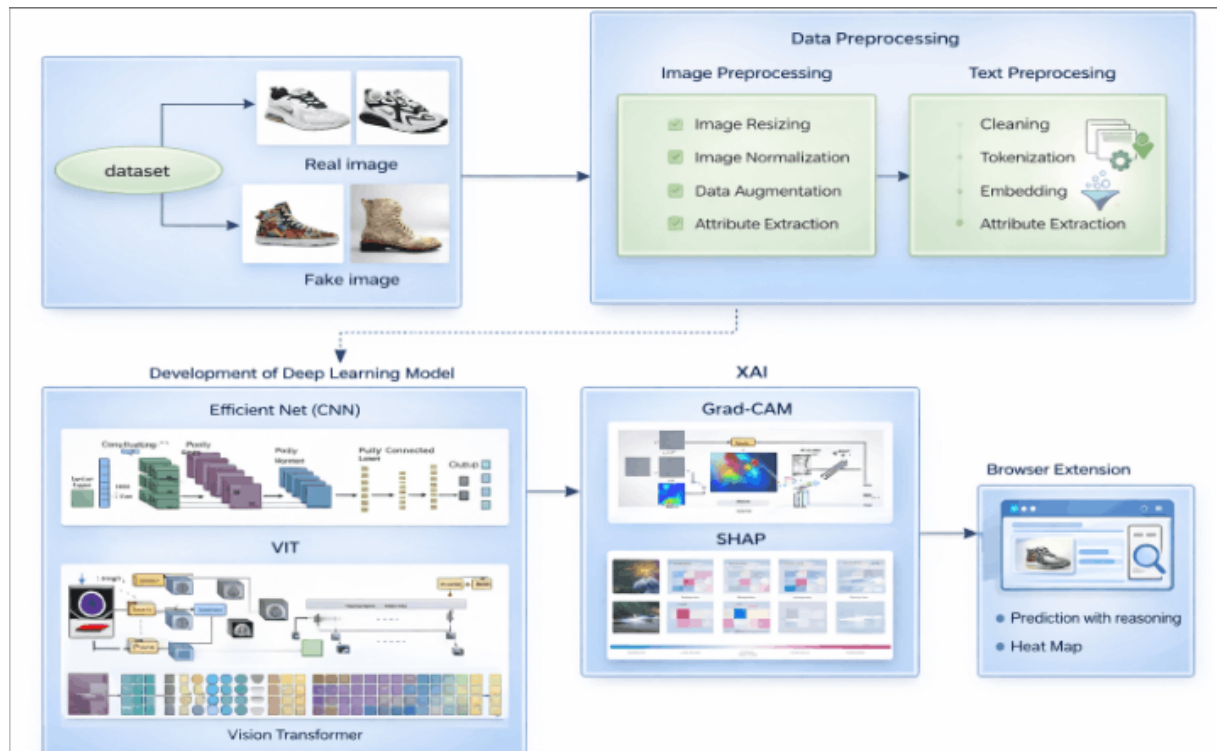


Figure 3.1: Architecture Diagram

- Sequence Diagram – This diagram shows how the user interacts with the browser extension and the backend models in real-time. It outlines the steps for extracting image or text from a webpage, sending that data for analysis, running the classification and explanation models, and then returning the final result to display to the user.

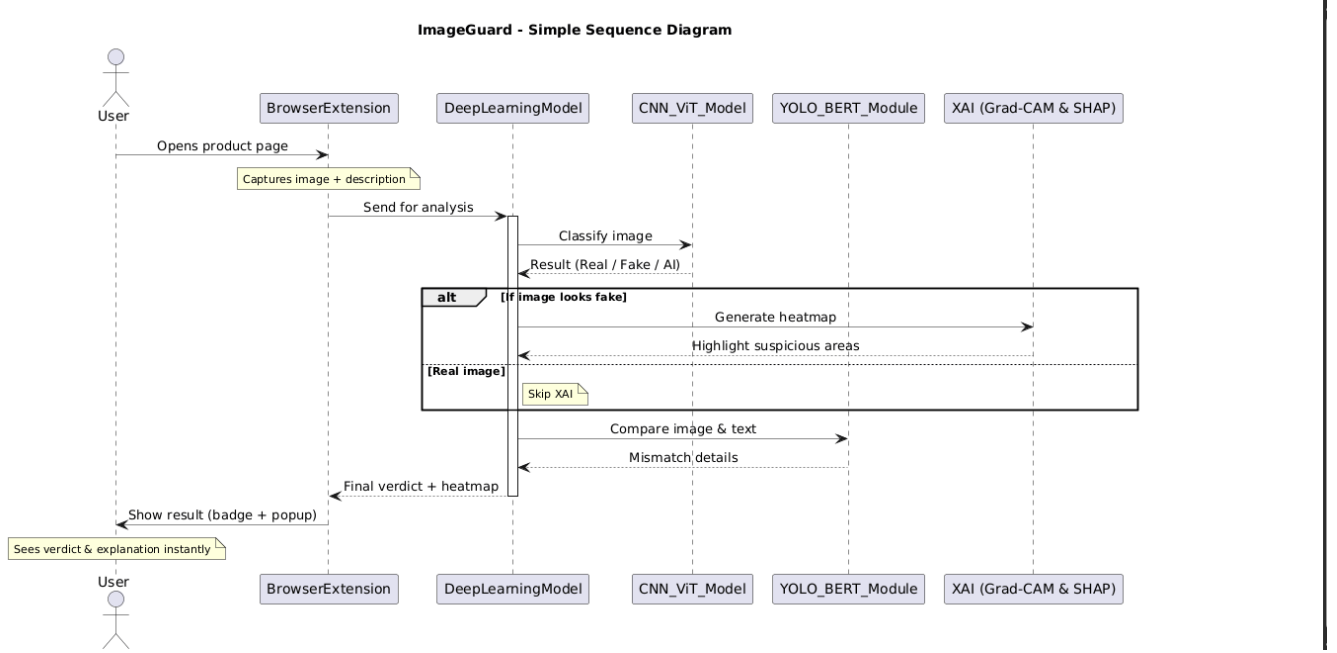


Figure 3.2: Sequence Diagram

3.3 Component Design

- Dataset Preparation Module:** Responsible for collecting a large, labeled dataset from sources like online marketplaces, benchmark datasets, and synthetic generation techniques such as GANs. The data goes through preprocessing steps, including resizing, normalization, duplicate removal, and augmentation to ensure robustness and balance.
- Image Manipulation Detection Module (CNN + ViT):** Serves as the core classification module that uses a hybrid deep learning approach:
 - CNN (EfficientNet):** Extracts pixel-level features and detects small-scale changes, such as uneven lighting or textures.
 - Vision Transformer (ViT):** This module examines the images by splitting the images into patches and processing these patches with the transformer encoder layer with self-attention mechanisms. This helps the model detect global context relationships and structural inconsistencies in the images. Such inconsistencies are usually found in AI-generated images.
- Object and Metadata Verification Module (YOLO + Regex):** Performs

multimodal verification to make sure visual and textual information match: This module checks the consistency between the image of the product and its description. Product attributes, which are brand, color, size, and price, are identified using regular expressions as well as keyword mapping, which are matched against the results of object detection and color analysis.

- **Explainable AI Module (Grad-CAM + SHAP):** Enhances transparency and user trust by visualizing model decisions:
 - **Grad-CAM:** Generates heatmaps from CNN layers to show which parts of the image affected the classification.
 - **SHAP:** Assigns importance scores to image areas based on game theory. It explains how these areas contribute to the model's output.
- **Browser Extension Integration Module:** Acts as the frontend interface built with JavaScript. It pulls product images and metadata from e-commerce pages. Then, it sends this information to the backend model for analysis. Finally, it shows the detection results to users through a badge, popup, or heatmap overlay.

3.4 Algorithms Used and Design

- Convolutional Neural Networks (CNNs): Extract pixel-level patterns such as edges, lighting, and texture inconsistencies.
- Vision Transformers (ViT): Split images into patches to analyze long-range dependencies and global relationships.
- YOLO (You Only Look Once): Performs real-time object detection to identify key objects in product images.
- Grad-CAM and SHAP: Used for explainability, highlighting suspicious image regions that influence predictions.

3.5 Software and Hardware Requirements

Software Requirements:

- Programming: Python 3.10 or higher
- Frameworks: PyTorch, TensorFlow, OpenCV, FastAPI
- Libraries: Grad-CAM, SHAP, timm (for ViT/EfficientNet), BeautifulSoup, Selenium (for scraping)
- Frontend: HTML, CSS, JavaScript (planned for browser extension)

Hardware Requirements:

- GPU-enabled machine: NVIDIA GPU or Google Colab Pro (recommended for training)
- Minimum 16 GB RAM and 100 GB storage
- Stable internet connection: Required for data scraping and potential cloud deployment

3.6 Dataset Identified

- Online Marketplaces: Collecting a large labeled dataset of real and fake product images from e-commerce sites.
- Synthetic Data: Utilizing images generated by AI (GAN/AI-generated images) to train the model against modern fake threats.
- Benchmark Datasets: Incorporating established benchmark datasets relevant to image manipulation detection.
- Varied Manipulation Types: Ensuring the dataset includes various fakes such as AI-generated, Photoshopped, and enhanced images.
- Text and Image Data: The dataset includes both images and corresponding product descriptions for multimodal analysis.

3.7 Work Breakdown and Responsibilities

The development of the ImageGuard project was a product of teamwork in which each team member was given different responsibilities based on the different modules used in the development of the system. This ensured efficiency in the development of the project.

Aleena P

The implementation of the Vision Transformer (ViT-Base) model for detecting fake and manipulated images was carried out by Aleena P. This entailed data preprocessing, model training, evaluation of the transformer architecture, and performance analysis of the model. In addition, Aleena P was also involved in creating the product description checking module that checks for consistency between product image and description.

Alen Biju Nilakkappillil

Alen Biju was responsible for the collection of the dataset for training the models and testing them. The dataset collection process included collecting real images of products and fake images of products. The dataset was then preprocessed for training the models. Alen was also responsible for the development of the Chrome extension, which would facilitate real-time interaction between the user and the system.

Anand Basil

Anand Basil, was responsible for the implementation of the Explainable AI component. He used the Grad-CAM and SHAP techniques for the visualization of the predictions made by the deep learning models. He was also responsible for the training of the models, as well as the backend integration for the communication of the browser extension with the deep learning models through the FastAPI server.

Anna Praveen

The development of the CNN-based model using EfficientNet-B2 was carried out by Anna Praveen. She worked on training and tuning the model for effective detection of fake images. She also worked on the verification module for the product description.

3.8 Project Timeline

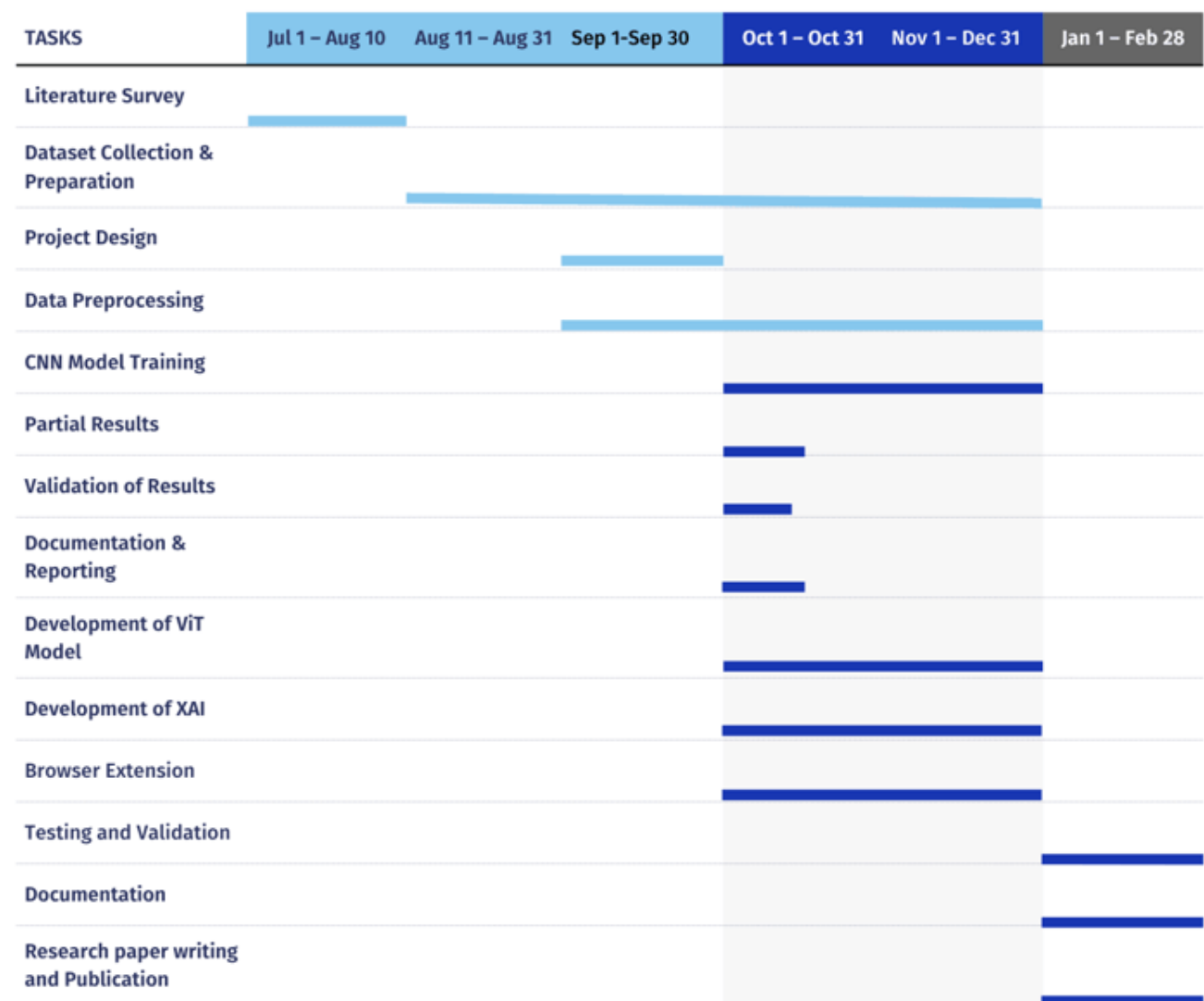


Figure 3.3: Project Timeline

3.9 Summary

This chapter laid the technical groundwork for developing ImageGuard. By combining CNN and ViT models with multimodal analysis, including YOLO and regex, and integrating Explainable AI tools like Grad-CAM and SHAP, the project aims to provide a solution that is accurate, clear, and easy to use against visual fraud in e-commerce. The design meets non-functional requirements of high accuracy and near real-time performance. This ensures that the final product will be a dependable tool for improving the

authenticity and trustworthiness of online marketplaces.

Chapter 4

System Implementation

4.1 Dataset Engineering and Preprocessing

The foundation of the system is a curated set of data that attempts to capture the diversity that exists in online marketplaces.

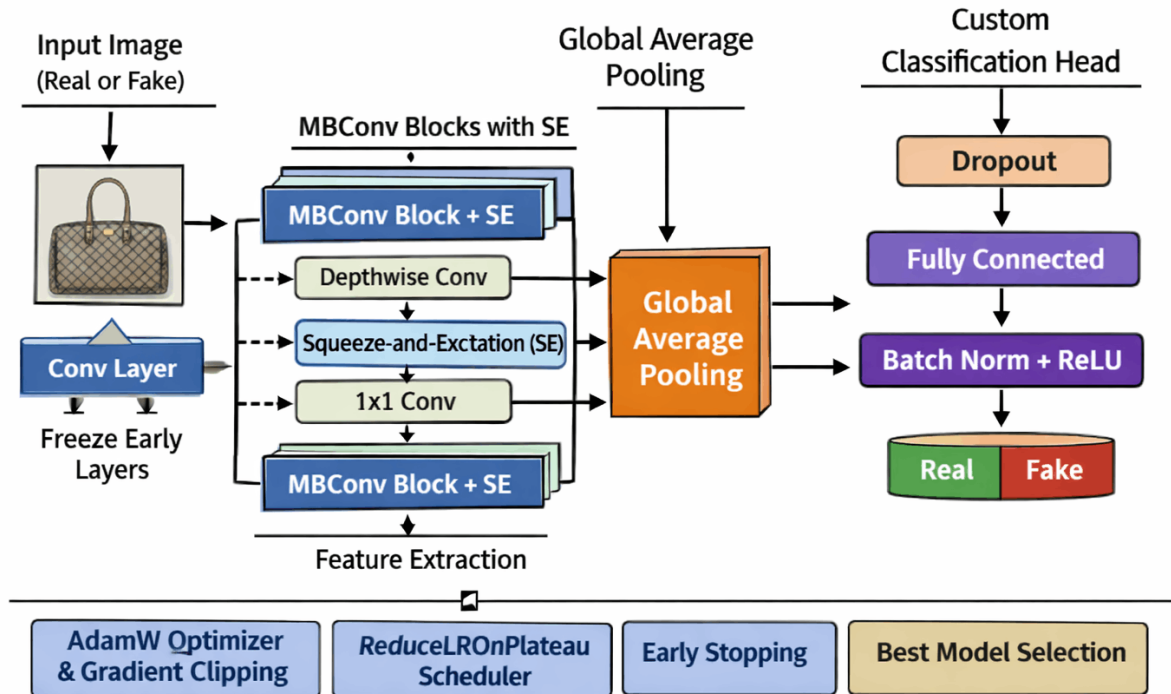
- **Data Acquisition:** The data set for this model comprises 19,887 images. The authentic data set was obtained from the original brand websites such as Nike, Rolex, Rado, Tissot...etc. The fake data set was created by using the CycleGAN and the FLUX.1 Dev AI generator for fine texture inconsistencies and artifacts.
- **Preprocessing Pipeline:** The images undergo a standard process in which all images are resized to a fixed resolution. The resolution is 224×224 pixels. Pixel values are normalized using mean values from ImageNet ($\mu = [0.485, 0.456, 0.406]$) and standard deviation values ($\sigma = [0.229, 0.224, 0.225]$) to keep the intensity distribution consistent.
- **Imbalance Mitigation:** To prevent the models from showing bias toward the dominant class, the system uses a weighted random sampler and calculates class-weighted loss functions based on the inverse class frequency.

4.2 Deep Learning Model Development

ImageGuard uses a hybrid vision approach by applying two different architectures to achieve the highest possible accuracy in the detection process.

- **EfficientNet-B2 (CNN):** This model is used due to its capacity to recognize local visual features such as material types (leather, fabric), logo shapes, stitching patterns, and others. The architecture uses MBConv (Mobile Inverted Bottleneck)

blocks with Squeeze-and-Excitation (SE) modules, which adjust channel importance to focus on the most informative features.



EfficientNet-B2 for Fake Image Detection

Figure 4.1: EfficientNet-B2 Architecture Diagram

- **Vision Transformer (ViT-Base):** While the CNN focuses on local details, the ViT model treats the image as a sequence of 16×16 patches. With a Multi-Head Self-Attention mechanism, it learns global contextual relationships that are crucial for identifying structural distortions and symmetry issues common in AI-generated images.

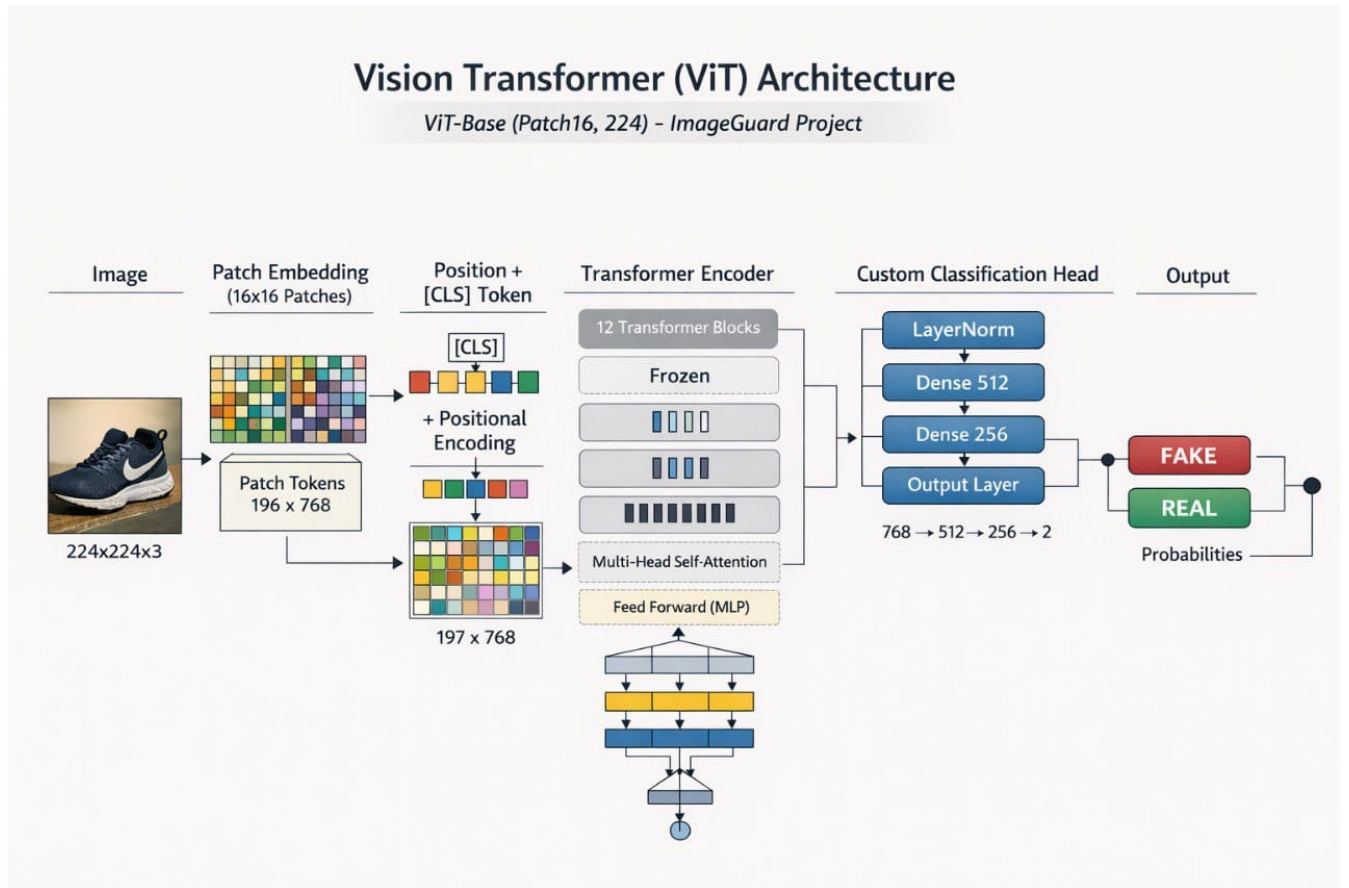


Figure 4.2: ViT Architecture Diagram

4.3 Multimodal Verification Layer

ImageGuard does more than just look at the pixels in an image. It also checks the product description to make sure it matches the image.

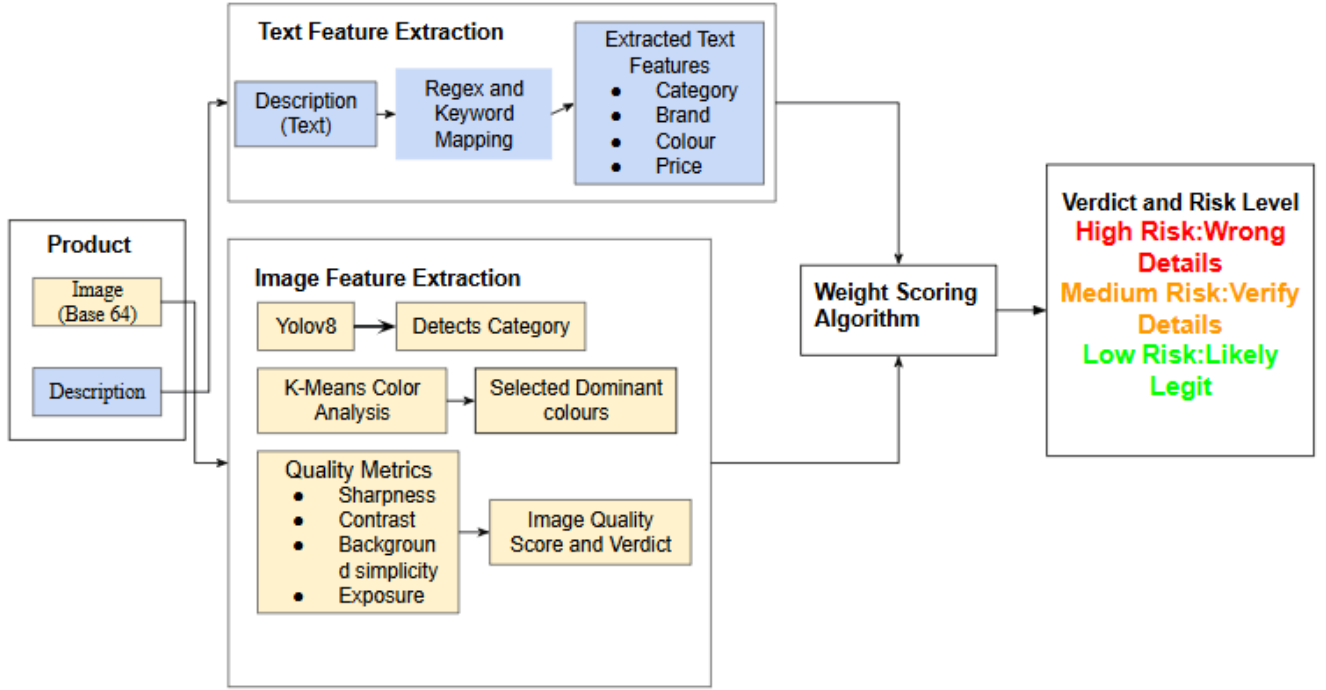


Figure 4.3: Product Description Analysis Architecture Diagram

- **Text Feature Extraction:** The system uses Regular Expressions (Regex) and keyword mapping to scrape and parse product titles and descriptions for “Ground Truth” attributes: Brand, Color, Size, Price, and Condition.
- **Visual-Text Cross-Referencing:** These attributes are compared to image data processed using a YOLOv8 classifier for category identification and K-Means Color Analysis.
- **Weighted Scoring Algorithm:** A risk-assessment engine calculates a Match Score (0 to 1) and a Risk Score. A category mismatch (e.g., text says “Watch” but the image is a “Shoe”) triggers a high-risk alert, penalizing the score and flagging the item as a potential scam.

4.4 Explainable AI (XAI) Integration

Transparency is integrated into the model’s decision-making process to build user trust.

- **Grad-CAM:** For the CNN model, Grad-CAM computes gradients of the predicted

class to highlight specific regions, such as blurred textures or misplaced logos, producing a visual heatmap for the user.

- **SHAP (SHapley Additive exPlanations):** SHAP is used to evaluate the contribution of features such as blurriness, edge sharpness, and compression artifacts. The system can use it to explain why an image has been classified based on quality metrics.

4.5 Real-Time Deployment Strategy

The entire system is deployed via a user-centric browser environment.

- **Browser Extension:** Developed as a Chrome Extension using HTML, CSS, and JavaScript, it captures images and descriptions from the webpage DOM without altering the marketplace infrastructure.
- **FastAPI Backend:** The extension interacts with the FastAPI backend through REST APIs. The backend runs EfficientNet and ViT models as inference services and sends the final classification, confidence level, and XAI reasoning to the user in real-time.

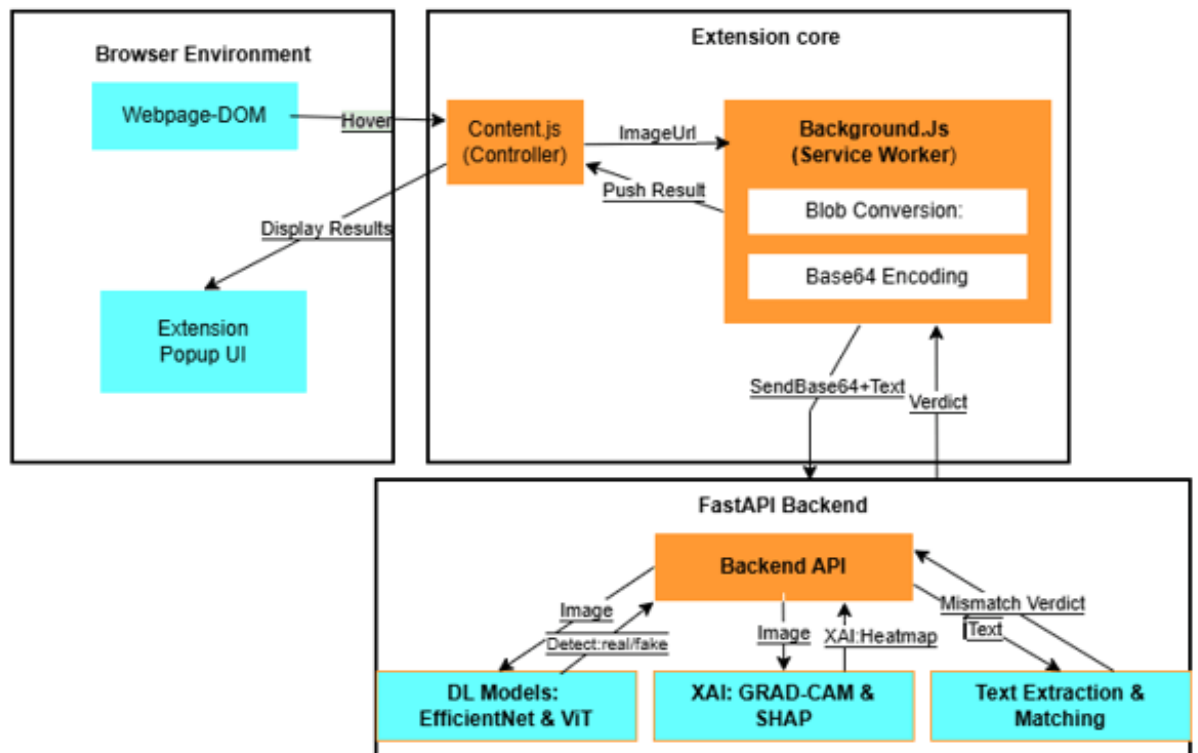


Figure 4.4: Browser Extension Workflow

Chapter 5

Results and Discussion

5.1 Experimental Results and Performance Evaluation

To assess the effectiveness of the proposed ImageGuard framework for fake image detection in online marketplaces, multiple standard classification and diagnostic metrics are employed. These metrics evaluate both predictive performance and decision reliability under real-world conditions.

5.1.1 Performance Metrics of Deep learning Models and Model Comparison

The framework was rigorously tested using a balanced test set of **3,217 images**, consisting of **1,500 fake** and **1,717 real** samples. Performance was assessed across two primary architectures: **EfficientNet-B2 (CNN)** and **Vision Transformer (ViT-Base)**.

Metric	EfficientNet (CNN)	Vision Transformer (ViT)
Test Accuracy	99.35%	99.01%
ROC AUC Score	0.9998	0.9992
Best Validation Accuracy	99.25%	98.66%
Best Validation Loss	0.0315	0.0694
Total Training Time	55.7 minutes	238.8 minutes

Table 5.1: Quantitative Performance Comparison of Deep Learning Models

The CNN model converged significantly faster than the ViT model, requiring only 55.7 minutes of training compared to the ViT’s 238.8 minutes. However, both architectures proved highly reliable, achieving near-perfect ROC AUC score.

5.1.2 Receiver Operating Characteristic (ROC) Curve Analysis

The Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) score were utilized to evaluate the discrimination capability of the developed models. These metrics represent the model's ability to correctly rank a randomly chosen fraudulent image higher than a real one.

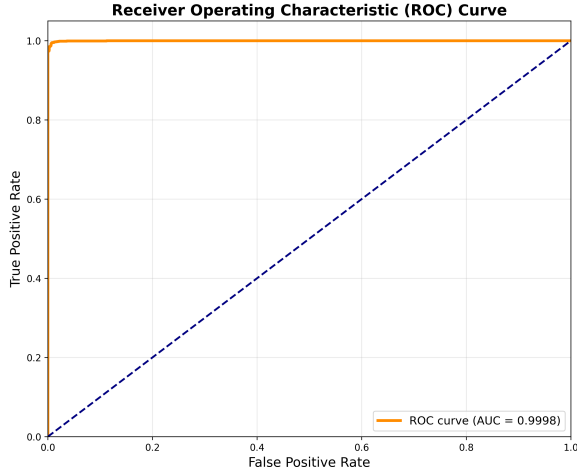


Figure 5.1: EfficientNet ROC Curve

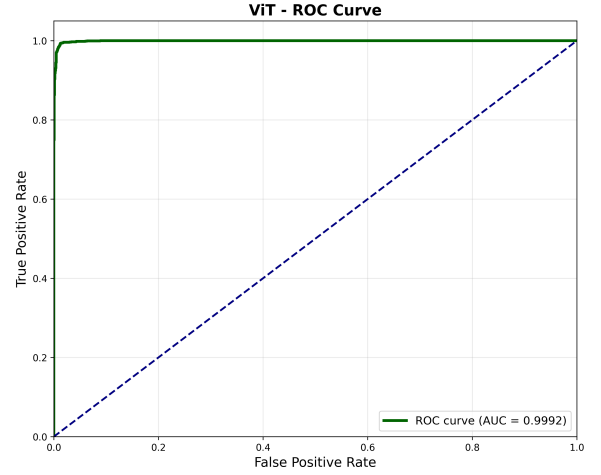


Figure 5.2: ViT ROC Curve

ROC and AUC Performance

The experimental evaluation demonstrated near-perfect separability for both architectures, confirming their reliability in differentiating between authentic and manipulated images.

- **EfficientNet (CNN) Results:** The CNN model achieved a ROC AUC score of **0.9998**. The ROC curve closely followed the top-left boundary, indicating an excellent trade-off between true positive and false positive rates at various classification thresholds.
- **Vision Transformer (ViT) Results:** The ViT model attained a ROC AUC score of **0.9992**. Similar to the CNN, the curve exhibited a sharp ascent toward the top-left limit, reflecting high discriminative power.

5.1.3 Training and Validation Performance

The models were trained on 13,456 samples and validated on 3,214 samples to ensure robust generalization.

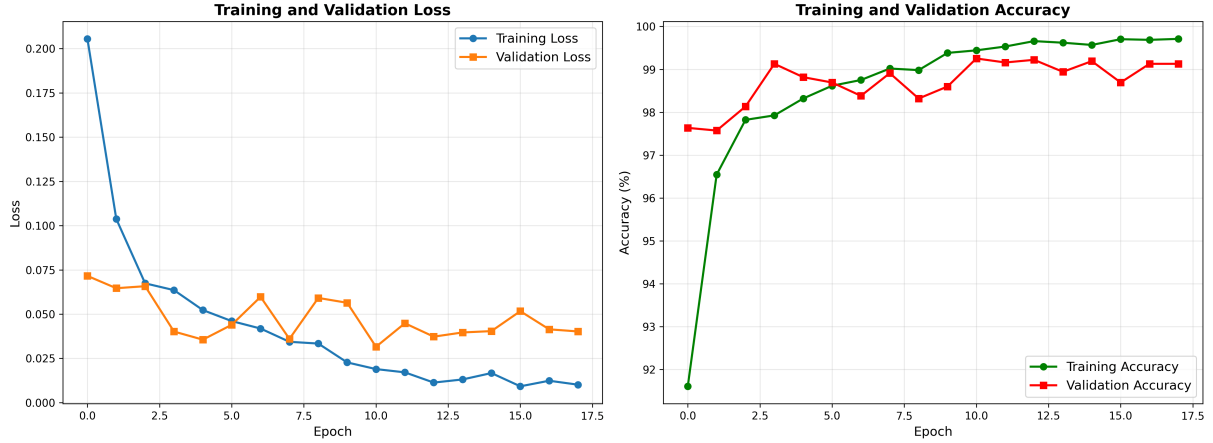


Figure 5.3: Training and Validation loss and Accuracy of EfficientNet

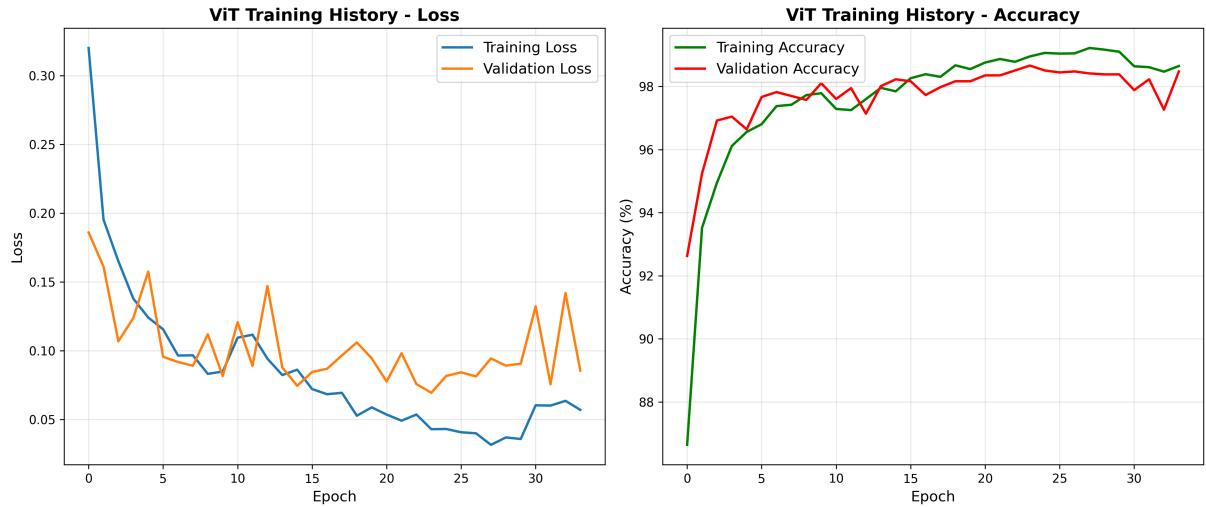


Figure 5.4: Training and Validation loss and Accuracy of ViT

- **EfficientNet Convergence:** The training loss decreased rapidly in early epochs, settling at a low value, which indicates effective optimization. It achieved a best validation accuracy of 99.25% and a validation loss of 0.0315.
- **ViT Convergence:** The ViT model showed stable convergence with training and validation curves closely aligned, reaching a peak validation accuracy of 98.66% and a minimum loss of 0.0694.

- **Consistency:** The close match between training and validation curves for both models confirms balanced learning dynamics and minimal overfitting.

5.1.4 Confusion Matrix Insights

The confusion matrices for both models demonstrate a high degree of class separability:

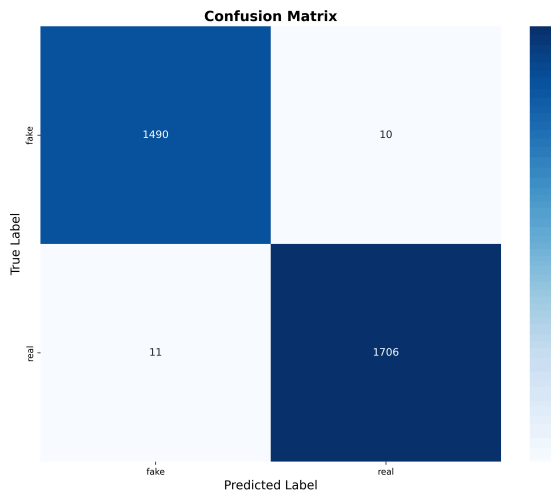


Figure 5.5: EfficientNet Confusion Matrix

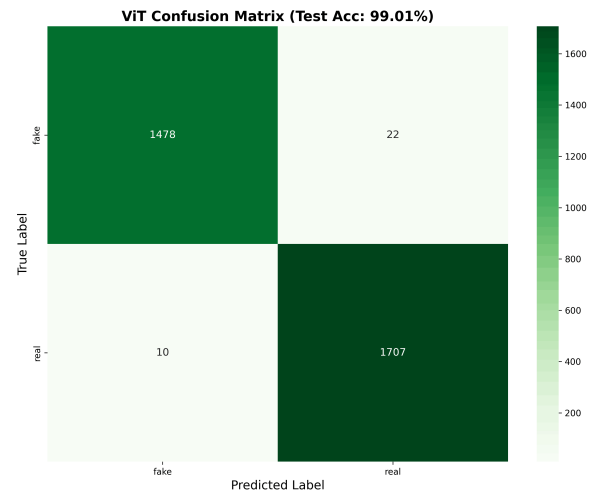


Figure 5.6: ViT Confusion Matrix

- **EfficientNet (CNN):** Correctly identified **1,490 fake images** and **1,706 real images**. Only 10 real images were misclassified as fake, and 11 fake images were misclassified as real.
- **Vision Transformer (ViT):** Correctly identified **1,478 fake images** and **1,707 real images**. The slight increase in misclassified fakes (22 images) compared to the CNN suggests its strength lies more in global structural evaluation than minute texture detection.

5.2 Explainable AI (XAI) Outcomes

ImageGuard provides transparency through visual and feature-level explanations to enhance user trust.

5.2.1 Visual Analysis (Grad-CAM)

- **CNN Insights:** Grad-CAM heatmaps highlight local suspicious regions, such as blurred textures, distorted logos, or AI-generated artifacts on product surfaces.

- **ViT Insights:** Patch-level analysis (using 16x16 patches) visualizes how the model attends to global relationships, capturing structural distortions and symmetry issues common in AI-generated products.

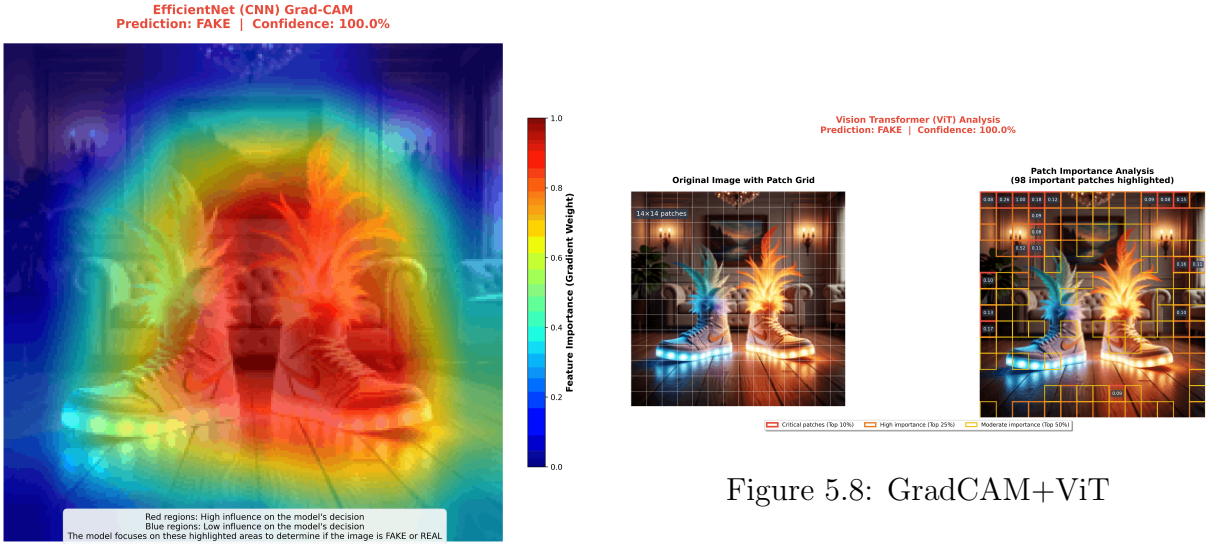


Figure 5.7: GradCAM+EfficientNet

5.2.2 Feature Contribution (SHAP)

SHAP analysis reveals the contribution of specific visual attributes to the final decision:

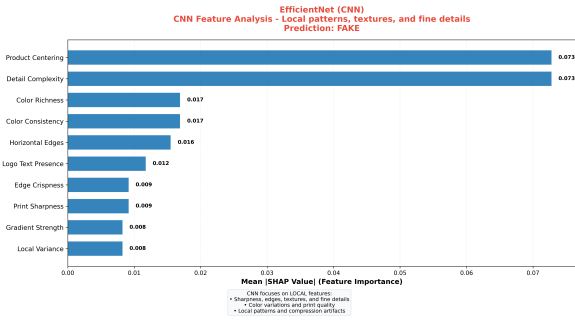


Figure 5.9: SHAP+EfficientNet

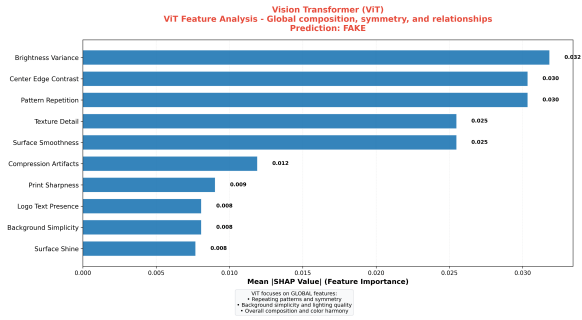


Figure 5.10: SHAP+ViT

- **CNN Analysis:** Ranked **Detail Complexity** (0.128 score) and **Blurriness** (0.071 score) as top features, indicating a focus on local image quality and sharpness.
- **ViT Analysis:** Identified **Regular Patterns** (0.078 score) and **Product Centering** (0.058 score) as critical features, confirming its specialization in global composition and symmetry.

5.3 Browser Extension and Real-Time Deployment Results

This phase of the evaluation involved testing the ImageGuard browser extension in a live web environment to determine its utility as a real-time fraud detection tool for end-users.

5.3.1 Deployment and Integration Performance

The system was successfully deployed as a Chrome extension, facilitating instant image and description verification without modifying the existing marketplace infrastructure.

- **Data Extraction:** The extension effectively utilized the webpage Document Object Model (DOM) to capture product images and textual descriptions upon user interaction.
- **Latency:** REST-based communication with the FastAPI backend ensured that preprocessing and model inference (EfficientNet and ViT) occurred with minimal latency, maintaining a seamless browsing experience.

5.3.2 Real-World Detection Outcomes

Testing across various e-commerce platforms and search engines demonstrated high reliability in real-world scenarios.

- **Fraudulent Image Detection:** When tested on AI-generated product images found via web search, the extension correctly identified the samples as “Fake” with 100% confidence scores from both EfficientNet and ViT models.
- **Authentic Image Verification:** Live testing on official e-commerce sites (e.g., Amazon) resulted in a verdict of “Detected real,” with 100% confidence and zero false positives observed during the trial.
- **Interactive Explainability:** A key outcome was the successful delivery of XAI visualizations within the extension popup. Users were provided with Grad-CAM heatmaps and SHAP feature analysis charts directly in the browser interface, allowing them to see exactly why a product was flagged as suspicious.

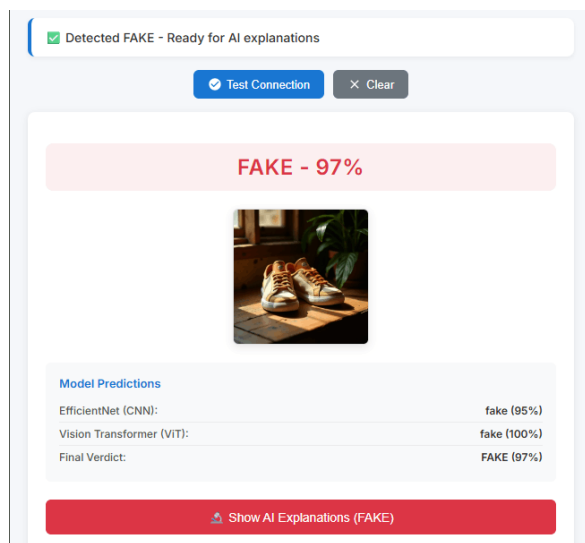


Figure 5.11: Detection of Fake image

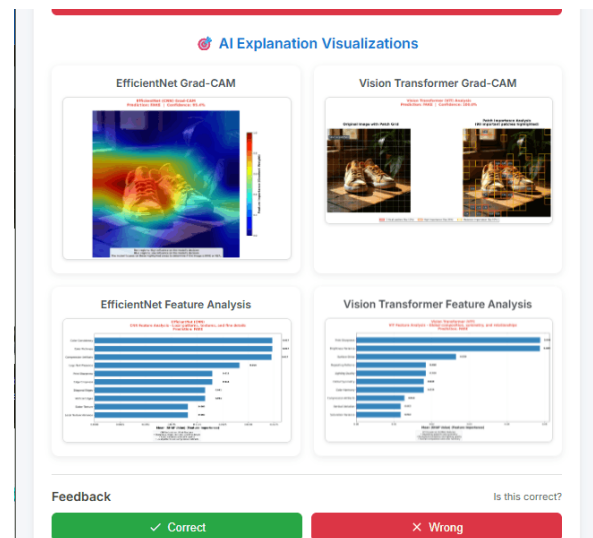


Figure 5.12: XAI explanation for fake image

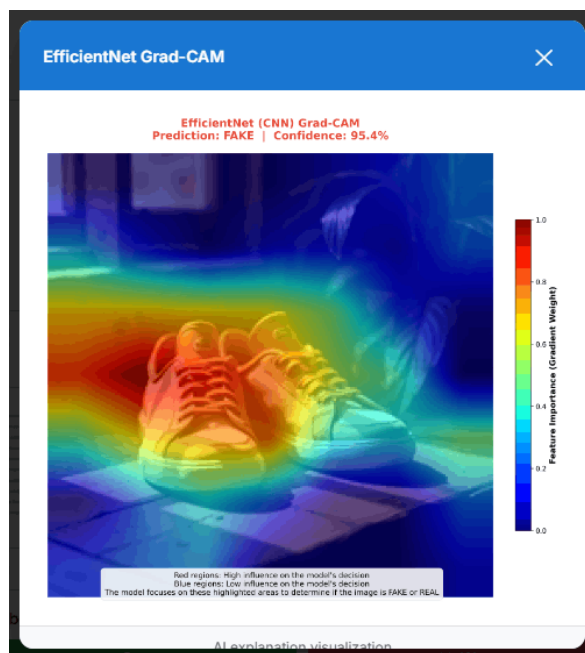


Figure 5.13: GradCAM+EfficientNet-Fake

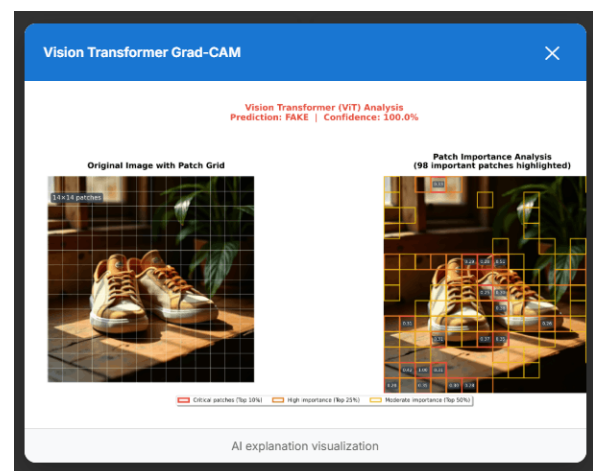


Figure 5.14: GradCAM+ViT-Fake

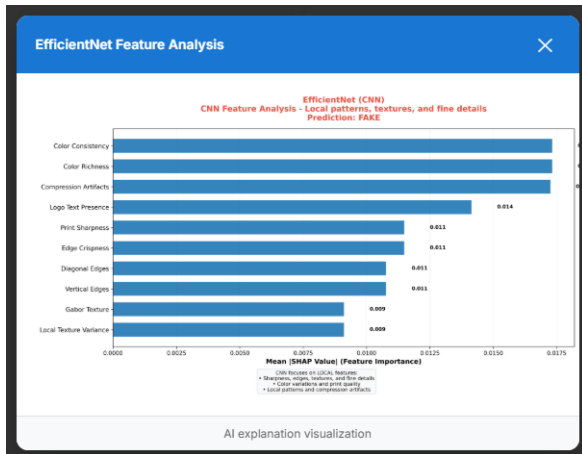


Figure 5.15: SHAP+EfficientNet-Fake

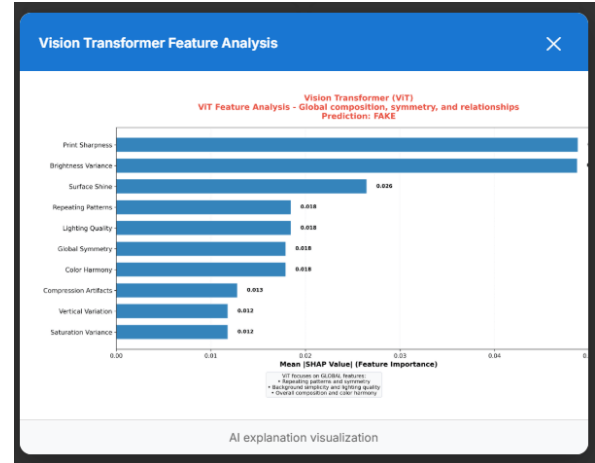


Figure 5.16: SHAP+ViT-Fake

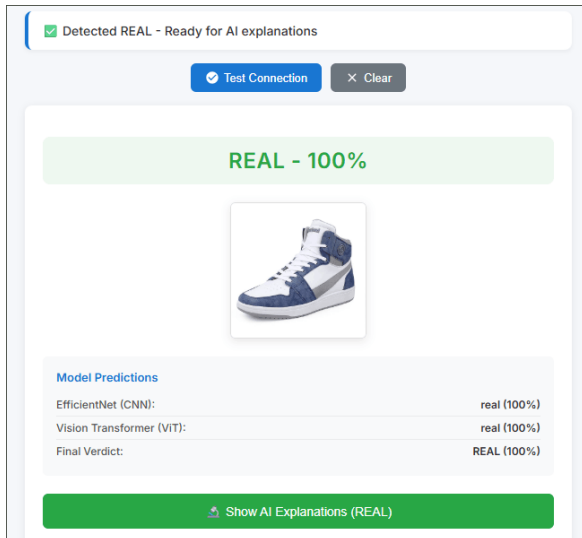


Figure 5.17: Detection of Real image



Figure 5.18: XAI explanation for real image

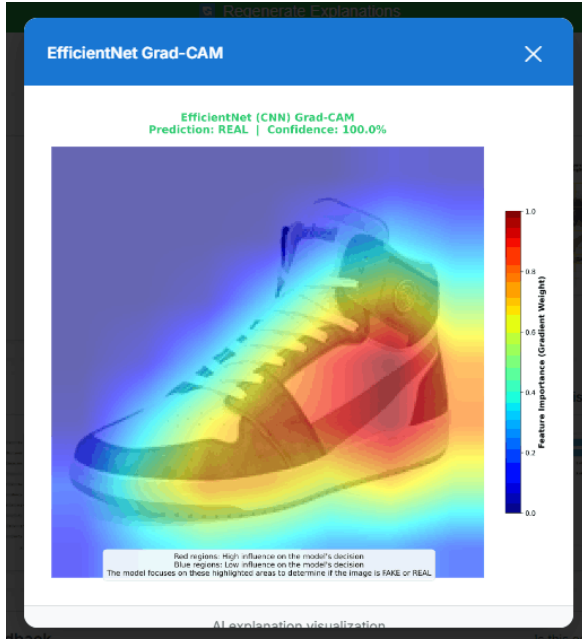


Figure 5.19: GradCAM+EfficientNet-Real

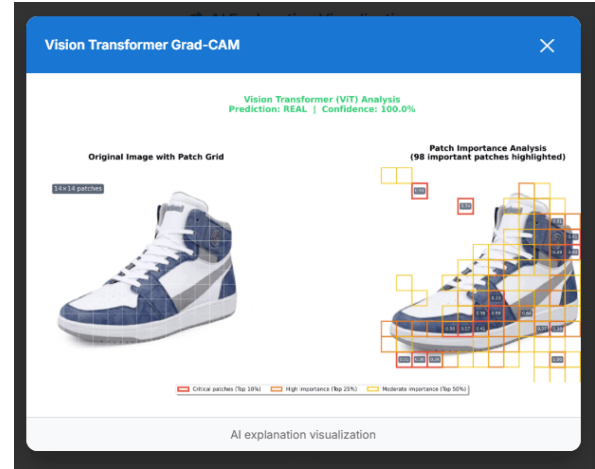


Figure 5.20: GradCAM+ViT-Real

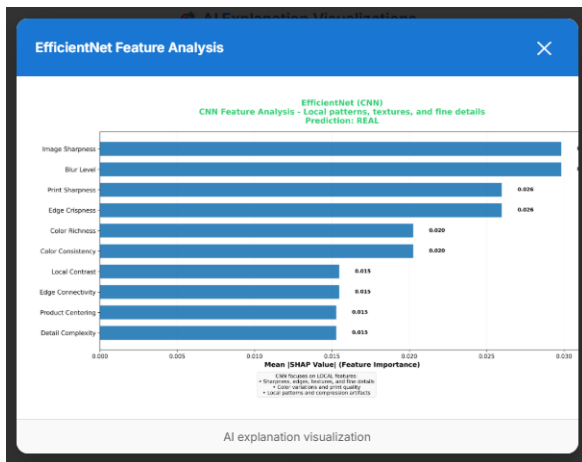


Figure 5.21: SHAP+EfficientNet-Real

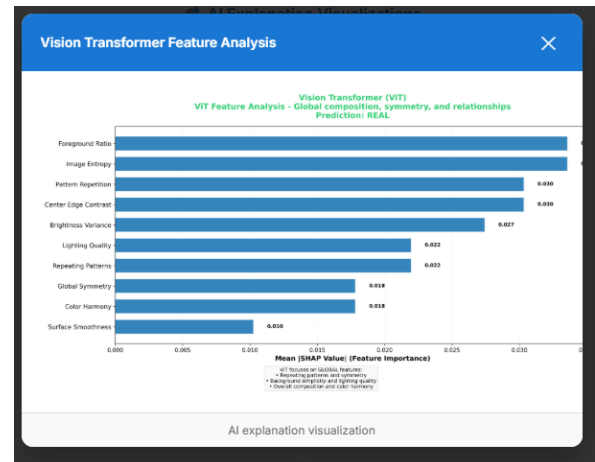


Figure 5.22: SHAP+ViT-Real

5.4 Multimodal Verification Results

The product description analysis module addresses “Bait & Switch” tactics by checking consistency between listing text and image metadata.

- **Likely Legitimate:** Listings with matching categories and colors achieved a Match Score of **1.00**, resulting in a “LOW RISK” verdict.
- **Potential Scam:** Listings where the text mentioned a “Watch” but the image

contained a “Shoe” resulted in a Match Score of **0.20** This triggered high-risk alerts such as “POTENTIAL SCAM” based on a Risk Score exceeding 60.

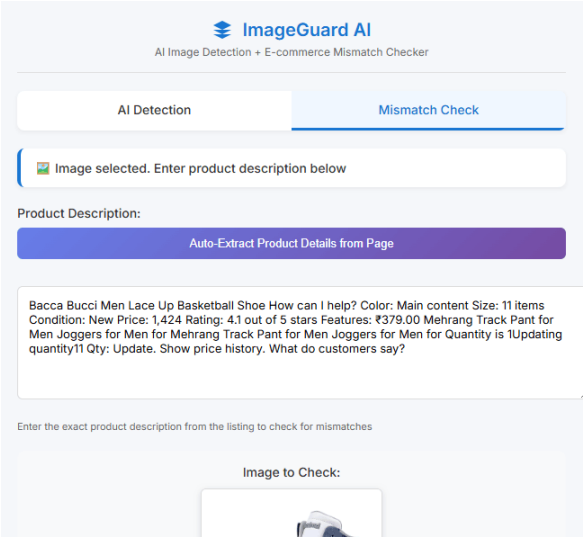


Figure 5.23: Product Description extracting

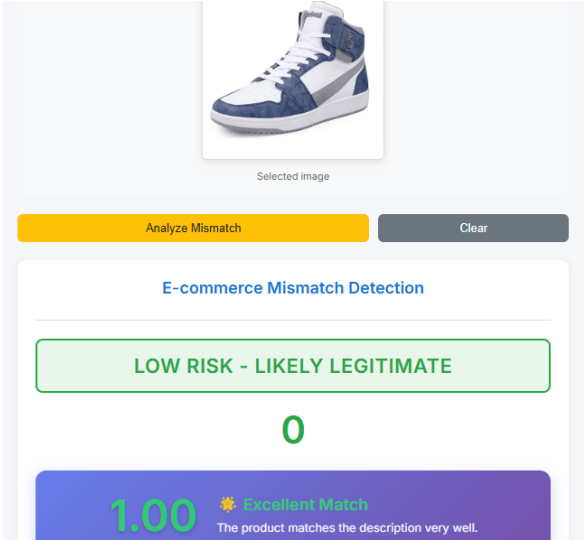


Figure 5.24: Product Description Analysis

5.5 Discussion

The experimental results confirm that a hybrid deep vision approach is superior for counterfeit detection. While the EfficientNet CNN provides rapid, high-accuracy verification of local textures, the ViT architecture is uniquely capable of identifying global structural inconsistencies in advanced AI-generated images. Furthermore, the multimodal verification layer serves as a critical safety net against scams that purely visual models might miss. The high ROC AUC scores and minimal misclassifications across both models establish ImageGuard as an effective tool for enhancing transparency and trust in online marketplaces.

Chapter 6

Conclusion and Future Scope

The ImageGuard project effectively tackles the growing issue of fake images in online marketplaces by creating a detailed detection system. By combining deep learning models, multimodal analysis, and explainable AI (XAI), the project provides a reliable tool for detecting fraud.

At the heart of this initiative is a deep learning model that uses both Convolutional Neural Networks (CNNs) and Vision Transformers (ViT). This model can accurately spot a wide variety of fake and altered images. The visual analysis is supported by a verification component that employs YOLO to identify objects in an image and regex to examine the related text. This ensures consistency between what the image shows and what the text describes.

A key feature of ImageGuard is its use of XAI techniques like Grad-CAM and SHAP. This goes beyond simply labeling an image as "real or fake." It provides clear, visual feedback about why an image was flagged. This transparency allows users and moderators to understand and trust the system's decisions. By making this technology accessible through an easy-to-use browser extension, ImageGuard empowers both consumers and platforms. Ultimately, the project significantly improves the authenticity and trustworthiness of online marketplaces, promoting a safer and clearer digital shopping experience.

Looking into the future of the ImageGuard system, it is possible that the dataset will be expanded to include more types of products and various examples of counterfeits to increase the accuracy of the image detection model. The system could be extended to various platforms, including mobile apps and online shopping sites. Retraining the image detection model using new data collected could also be an option to make the system more adaptable to new image manipulation techniques. Another possible extension of the system could be the detection of AI-generated images and improving the system's speed.

References

- [1] N. M. AlShariah, A. Khader, and J. Saudagar, “Detecting fake images on social media using machine learning,” *International Journal of Advanced Computer Science and Applications*, vol. 10, no. 12, pp. 170–176, 2019.
- [2] E. Jain and D. Kundra, “Efficientnet-based deepfake detection: A robust approach for real and fake media classification,” in *2024 Global Conference on Communications and Information Technologies (GCCIT)*. IEEE, 2024, pp. 1–6.
- [3] R. B. Reddy, T. N. Siva, B. S. Ram, K. R. Sree, and B. Suvarna, “Enhanced deep fake image detection via feature fusion of efficientnet, xception, and resnet models,” in *2025 6th International Conference on Mobile Computing and Sustainable Informatics (ICMCSI)*. IEEE, January 2025, pp. 1547–1552.
- [4] B. Ghita, I. Kuzminykh, A. Usama, T. Bakhshi, and J. Marchang, “Deepfake image detection using vision transformer models,” in *2024 IEEE International Black Sea Conference on Communications and Networking (BlackSeaCom)*. IEEE, June 2024, pp. 332–335.
- [5] C. V. Zyl, X. Ye, and R. Naidoo, “Harnessing explainable artificial intelligence for feature selection in time series energy forecasting: A comparative analysis of grad-cam and shap,” *Applied Energy*, vol. 353, p. 122079, 2024.
- [6] W. H. Abir, F. R. Khanam, K. N. Alam, M. Hadjouni, H. Elmannai, S. Bourouis, R. Dey, and M. M. Khan, “Detecting deepfake images using deep learning techniques and explainable ai methods,” *Intelligent Automation & Soft Computing*, vol. 35, no. 2, pp. 2151–2169, 2023.
- [7] R. Dwivedi, D. Dave, H. Naik, S. Singhal, R. Omer, P. Patel, B. Qian, Z. Wen, T. Shah, G. Morgan, and R. Ranjan, “Explainable ai (xai): Core ideas, techniques, and solutions,” *ACM Computing Surveys*, vol. 55, no. 9, pp. 1–33, 2023. [Online]. Available: <https://doi.org/10.1145/3532108>

- [8] M. Štancel and M. Hulič, “An introduction to image classification and object detection using yolo detector,” in *CEUR Workshop Proceedings*, vol. 2403, 2019, pp. 1–8.
- [9] C. Liu, Y. Tao, J. Liang, K. Li, and Y. Chen, “Object detection based on yolo network,” in *2018 IEEE 4th Information Technology and Mechatronics Engineering Conference (ITOEC)*. IEEE, Dec. 2018, pp. 799–803.
- [10] J. M. Iqbal, T. Manikandan, K. S. Tamilselvan, A. Muruganandham, and S. Karthikeyan, “Bert-based deep learning models for enhanced object detection in images,” in *2024 Second International Conference on Advances in Information Technology (ICAIT)*, vol. 1. IEEE, July 2024, pp. 1–6.

List of Publications

1. "ImageGuard: An Explainable Hybrid Deep Vision Framework for Counterfeit Image Detection in E-Commerce Marketplaces", Communicated to Ceylon Journal of Science-Scopus Indexed.

Appendix A: Presentation

ImageGuard - Detection of Fake Images in Online Marketplaces Using Deep Learning and Explainable AI

PROJECT PRESENTATION

Guide: Dr. Preetha K G

Team Members:

Aleena P
Alen Biju Nilakkappillil
Anand Basil
Anna Praveen

CONTENTS

- Introduction
- Problem Statement
- Implementation Scope
- Literature Survey
- Architecture Diagram
- Sequence Diagram
- Module Division
- Role of Each Individual
- Hardware and Software Requirements
- Project Schedule
- Results
- Assumptions
- Risk and Challenges
- Conclusion
- Future Plans
- References

INTRODUCTION

- Online marketplaces heavily rely on product images to attract buyers.
- Rise in **fake, manipulated, and AI-generated** images misleads consumers.
- Fake images cause financial loss, trust issues, and platform credibility damage.
- Manual image verification is **time-consuming and not scalable**.
- **Deep Learning** enables automatic detection of image forgery.
- Lack of transparency in DL models creates trust and explainability concerns.
- ImageGuard integrates **Deep Learning + Explainable AI (XAI)** Provides accurate detection along with visual explanations of decisions.
- Available as a **browser extension** to promote safe and transparent online shopping.

3

PROBLEM STATEMENT

- To **design and develop** a deep learning–based model for detecting **fake product images** in online marketplaces.
- The system will analyze visual patterns, product description inconsistencies, and image manipulations to differentiate between authentic and fake/AI-generated images.

4

IMPLEMENTATION SCOPE

1. **Model Development:** Build and train CNN (EfficientNet) and ViT models to classify images as real or fake.
2. **Explainability:** Implement Grad-CAM and SHAP to visually explain why an image is flagged as fake.
3. **Multimodal Verification:** Integrate YOLO and Regex to check consistency between product images and descriptions.
4. **Deployment:** Deliver real-time detection via an inference API and a lightweight browser extension.

5

Literature Review

Paper Title	Methodology	Strengths	Limitations
Detecting Fake Images on Social Media using Machine Learning [1]	CNN extracts features using convolutional layers, pooling, and activation functions	Learns visual patterns automatically, high accuracy in image classification	Needs large datasets and GPU support
Deepfake Image Detection using Vision Transformer Models [2]	ViT patches the image, encodes the position of each patch, and applies multi-head attention and MLP layers	Understands full image context accurately, better generalization	Requires more data, high computational cost
Harnessing eXplainable artificial intelligence for feature selection in time series energy forecasting: A comparative analysis of Grad-CAM and SHAP [3]	XAI analyzes data by making model predictions transparent and comprehensible	Provides clear visual explanations for model decisions	Adds extra steps and complexity to the system

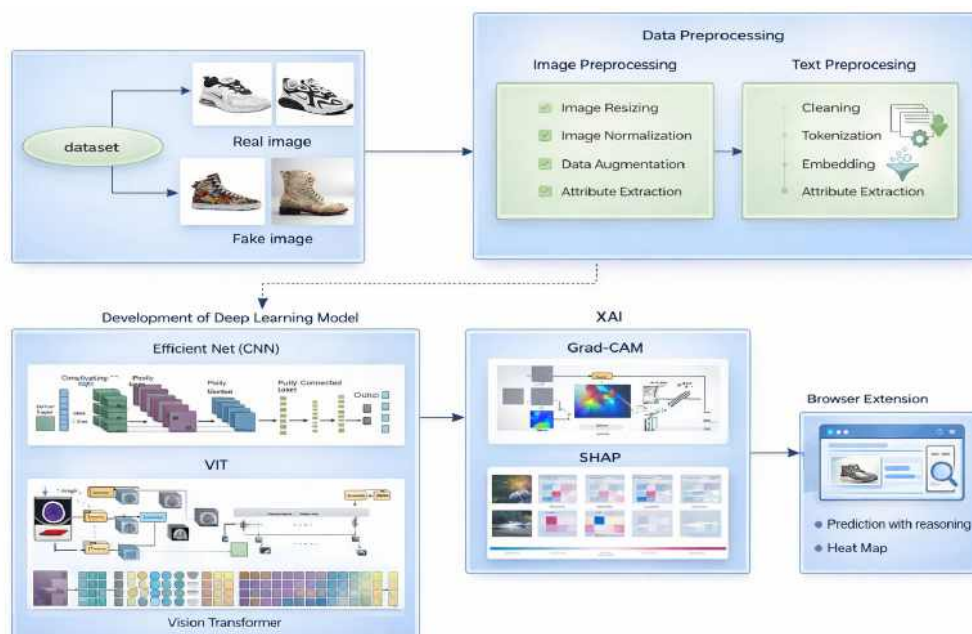
6

Literature Review

Paper Title	Methodology	Strengths	Limitations
An Introduction to Image Classification and Object Detection using YOLO Detector [4]	YOLO divides images into a grid, predicting bounding boxes and class labels	Fast and accurate object detection in one step	May miss small or overlapping objects
Fake Image Detection Using Deep Learning [5]	EfficientNet uses a compound scaling method to uniformly scale depth, width, and input resolution to achieve better performance with optimal efficiency.	EfficientNet provides high classification accuracy with fewer parameters and lower computational cost compared to traditional deep CNNs.	EfficientNet requires careful hyperparameter tuning and higher computational resources for larger variants like B4–B7

7

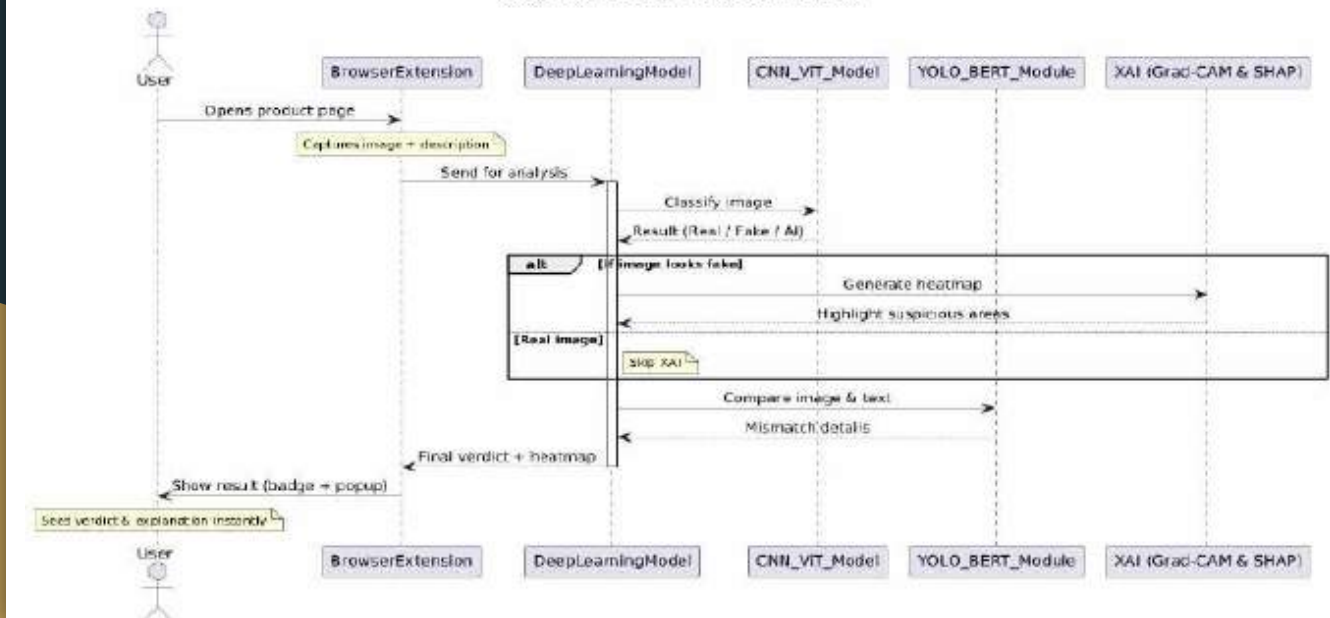
OVERALL ARCHITECTURE DIAGRAM



8

SEQUENCE DIAGRAM

ImageGuard - Simple Sequence Diagram



9

MODULE DIVISION

1. Dataset Preparation
2. Development of Deep Learning Model (CNN)
3. Development of Deep Learning Model (ViT)
4. Explainable AI (Grad-CAM + SHAP)
5. Product Description Analysis
6. Browser Extension (Frontend Integration)

10

ROLE OF EACH INDIVIDUAL

Alen Biju: Data collection and preprocessing, Chrome extension development and integration.

Anand Basil: Explainable AI (Grad-CAM, SHAP), Model training, Backend integration.

Aleena P: Vision Transformer (ViT-Base) implementation, Product description checking module.

Anna Praveen: CNN development (EfficientNet-B2), Dataset preparation, Product description verification.

11

HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements

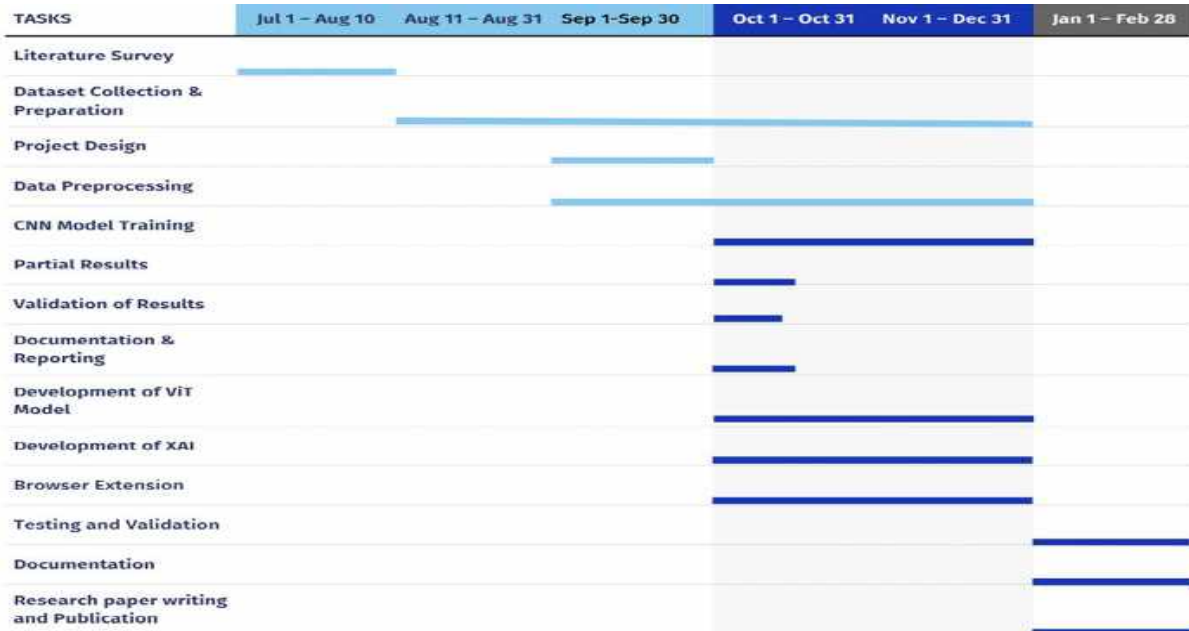
- System with minimum Intel i5 / Ryzen 5 processor
- At least 8GB RAM (16GB recommended for training)
- NVIDIA GPU (4GB+ VRAM recommended) for model training
- Minimum 20GB free storage for dataset and models

Software Requirements

- Frontend: HTML, CSS, JavaScript, Chrome Extension
- Backend: Python 3.8+, FastAPI (Python)
- Machine Learning: PyTorch, timm (EfficientNet-B2, ViT-Base), SHAP library, Grad-CAM
- Dataset: Manually prepared

12

PROJECT SCHEDULE



13

1. DATASET PREPARATION

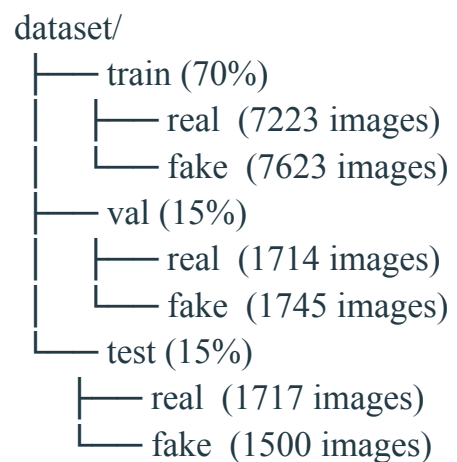
- Collection of authentic and fake / AI-generated product images
- Data sourced from online marketplaces and synthetic generation tools (CycleGAN)
- Labeling of images as Real or Fake
- Removal of noisy, duplicate, and low-quality samples
- Dataset split into training, validation, and testing sets

Shoes:- Real: 3757 ; Fake: 5199

Watches:- Real: 6897 ; Fake 5669

Real Images: 10654 images

Fake Images: 10868 images



14

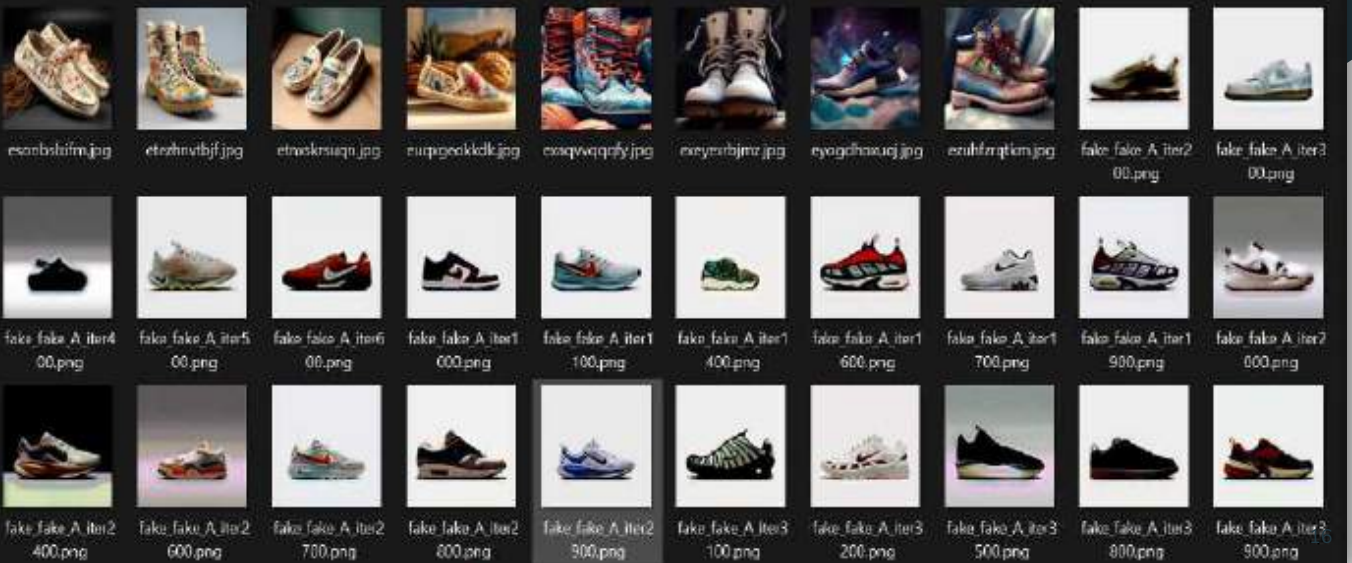
1. DATASET PREPARATION

Dataset Creation - Real



1. DATASET PREPARATION

Dataset Creation - Fake



2. DEVELOPMENT OF DEEP LEARNING MODEL (CNN)

- EfficientNet-B2 is a convolutional neural network designed to achieve high accuracy with computational efficiency.
- It uses convolution operations to extract spatial features from images.
- The model applies compound scaling, which balances network depth, width, and input resolution for optimal performance.
- EfficientNet focuses on learning local patterns such as edges, textures, and fine details, making it effective for visual recognition tasks.
- It enables efficient feature extraction with fewer parameters, allowing detection of subtle texture inconsistencies while maintaining fast inference speed.

17

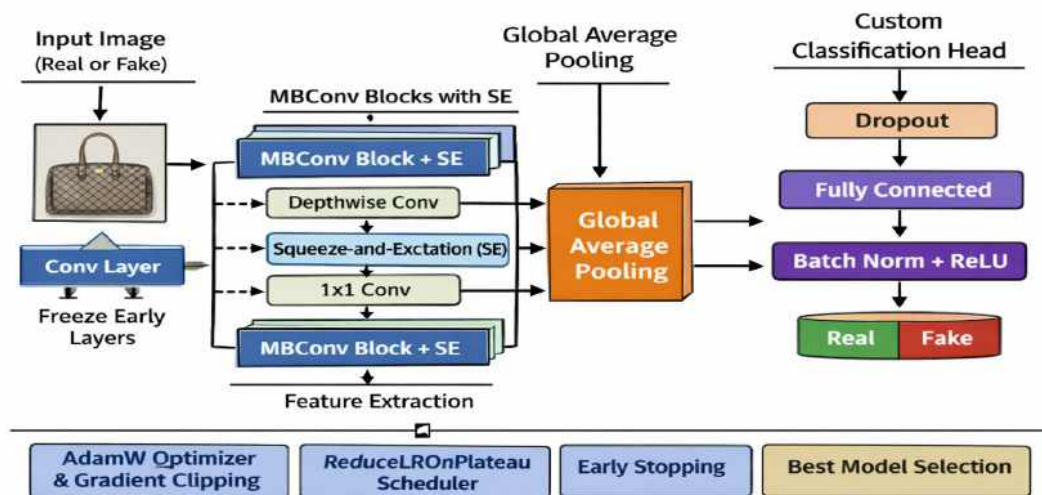
2. DEVELOPMENT OF DEEP LEARNING MODEL (CNN)

Working of EfficientNet-B2

- The input image is first resized to a fixed resolution and passed through an initial convolution layer that captures basic visual features such as edges and color gradients.
- The feature maps are then processed through multiple MBConv (Mobile Inverted Bottleneck) blocks, which use depthwise separable convolutions to efficiently learn spatial and channel-wise patterns while reducing computational cost.
- Within these blocks, Squeeze-and-Excitation (SE) modules adaptively recalibrate channel importance, allowing the network to focus on the most informative features.
- After feature extraction, a global average pooling layer converts the spatial feature maps into a compact feature vector representing the entire image.
- Finally, the feature vector is passed through a fully connected classification layer, enabling the model to make predictions based on the learned hierarchical representations.

18

2. DEVELOPMENT OF DEEP LEARNING MODEL (CNN)



EfficientNet-B2 for Fake Image Detection

19

3. DEVELOPMENT OF DEEP LEARNING MODEL (ViT)

- Vision Transformer (ViT) applies the **Transformer architecture** (originally from NLP) to images.
- Instead of convolutions, it treats image as a sequence of patches and uses self-attention mechanism.
- Vision Transformers enable global context learning, allowing them to capture structural inconsistencies, effectively detect AI-generated images, and model relationships between objects more accurately.
- The Vision Transformer works by resizing an image, splitting it into fixed-size patches, flattening them into vectors, converting them into embeddings, and processing them through transformer encoder layers to learn global relationships.

20

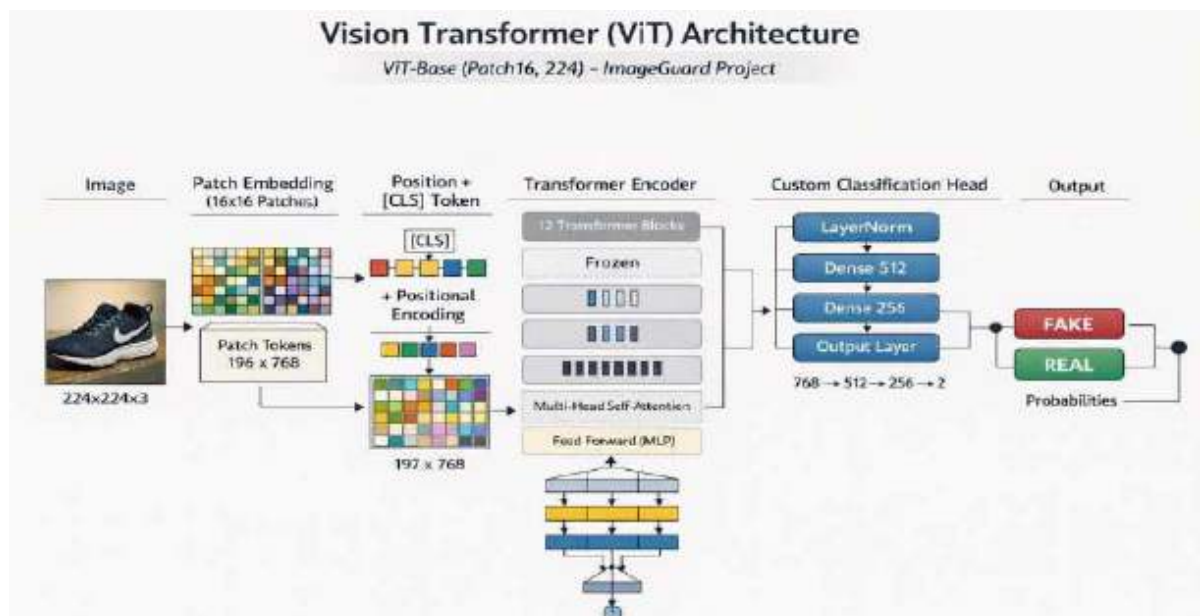
3. DEVELOPMENT OF DEEP LEARNING MODEL (ViT)

Working of ViT

- The input image is divided into small patches (e.g., 16×16), and each patch is flattened and converted into a vector representation called a patch embedding.
- Positional encoding is added to these embeddings so the model can understand the spatial order and location of each patch.
- A special learnable **CLS token** is appended to the patch sequence to aggregate information for final image classification.
- The sequence is passed through transformer encoder blocks consisting of Multi-Head Self-Attention, Feed Forward Networks, Layer Normalization, and Residual Connections to learn rich feature representations.
- Through the self-attention mechanism, each patch attends to all other patches, enabling the model to capture global relationships, long-range dependencies, and overall image context.

21

3. DEVELOPMENT OF DEEP LEARNING MODEL (ViT)



22

4. Explainable AI (GRAD-CAM & SHAP)

- Explainable AI (XAI) is used to understand why an image is classified as Real or Fake.
- It improves transparency, trust, and interpretability of the system.
- ImageGuard uses:
 - **Grad-CAM:** Highlights important image regions influencing the CNN (EfficientNet) prediction.
 - **ViT Patch Visualization:** Shows important image patches influencing the Vision Transformer decision.
 - **SHAP** (SHapley Additive Explanations): Analyzes feature contributions.

23

4.1 GRAD-CAM

- Visualize regions of the image that influence the CNN's decision.
- Helps identify suspicious areas in fake images.

Working

- The input image is processed by the EfficientNet CNN model, which predicts whether the image is **Real or Fake**.
- Grad-CAM analyzes the **final convolutional layer** of the model to determine which image regions contributed most to the prediction.
- These important regions are converted into a **heatmap** and **overlaid on the original image**.

Color Interpretation

- Red regions: Highest influence on the model's decision
- Yellow/Green regions: Moderate influence
- Blue regions: Low influence

24

4.2 Vision Transformer (ViT) Patch Visualization

- Vision Transformers divide an image into small patches.
- Patch visualization highlights which patches of the image influenced the model's prediction.

Working

- The input image is divided into **fixed-size patches**.
- Each patch is processed by the **Vision Transformer model**, which analyzes relationships between image regions.
- The model calculates **importance scores for each patch** based on its contribution to the prediction.
- Important patches are highlighted as a **grid overlay on the original image**.

Color Interpretation

- **Red patches:** Critical patches with highest importance
- **Orange patches:** High importance regions
- **Yellow patches:** Moderate importance regions
- Other patches are less influential and not highlighted.

25

4.3 SHAP

- Used to measure how different visual features influence the model's prediction.
- Helps identify which image characteristics support Real or Fake classification.
- Enables comparison of decision behavior between the CNN and ViT models.
- Image Features Used by SHAP include Blurriness level, Edge sharpness, Texture consistency etc.

SHAP Explanation Process

- **Feature Extraction:** Visual features are extracted from the input image.
- **Model Evaluation:** The CNN and ViT predictions are analyzed using these extracted features.
- **Contribution Calculation:** SHAP assigns an importance value to each feature based on its influence on the prediction.

Interpretation of SHAP Values

- Higher SHAP value → stronger influence on the prediction
- Lower SHAP value → weaker influence
- Important features are then ranked and visualized to explain the model's decision.

26

4.3 SHAP

SHAP for CNN

- Focuses on local texture-based features.
- Helps analyze features such as:
 - Blurriness levels
 - Edge sharpness
 - Texture consistency
- Explains how local visual details influence the model's decision.

SHAP for ViT

- Focuses on global image patterns and patch-level relationships.
- Helps analyze features such as:
 - Shape consistency
 - Structural patterns
 - Lighting and shading distribution
- Effective at identifying global inconsistencies often present in AI-generated images.

27

5. PRODUCT DESCRIPTION ANALYSIS

The Weight Scoring Algorithm

The system evaluates each listing using two metrics:

- **Match Score** – indicates how well the image matches the description
- **Risk Score** – indicates the level of potential mismatch or fraud

The algorithm starts with a neutral Match Score of 0.5 and a Risk Score of 0. It then adjusts these based on weighted priorities:

- **Category Match (40% Weight)**: The most critical check. A match adds **+0.3** to the score. A mismatch (Bait & Switch) penalizes the score by **-0.4** and adds **+40** to the Risk Score.
- **Color Consistency (30% Weight)**: Compares description keywords against HSV-mapped color clusters. An exact match adds **+0.25**, while a mismatch adds **+25** to the Risk Score.
- **Trust Signals (30% Weight)**: The presence of professional metadata like **Brand (+0.1)**, **Price (+0.1)**, and **Size (+0.1)** rewards the listing for being detailed and transparent.

28

5. PRODUCT DESCRIPTION ANALYSIS

- This module serves as the **Multimodal Verification Layer** of the ImageGuard system. Its purpose is to verify whether the product description matches the product image.
- The browser extension automatically scrapes the product title and description from the marketplace. This information is sent to the **FastAPI** backend, where features are extracted and a **match score** is calculated.

Technical Implementation Pipeline

The system performs analysis in three stages:

1. **Feature Extraction (NLP)**: Regular Expressions (Regex) and keyword mapping are used to extract key attributes from the product description, such as: Brand, Color, Size, Price, and Condition.
2. **Visual-Text Cross-Referencing**: These extracted text attributes are compared against the live outputs of the YOLOv8 classifier and K-Means Color Analysis.
3. **Similarity Logic**: The system applies a Weighted Scoring Algorithm where critical mismatches—such as a category conflict (text says "Watch" but image shows "Shoe")—trigger the highest risk alerts.

29

5. PRODUCT DESCRIPTION ANALYSIS

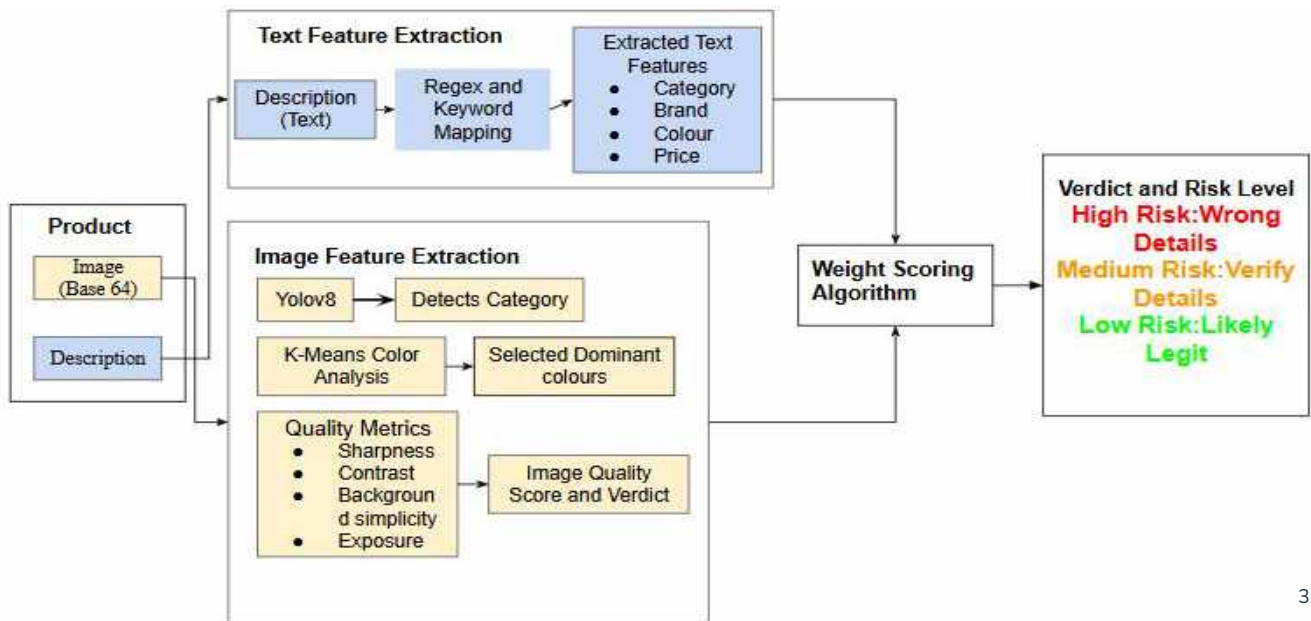
Risk Assessment & Final Verdicts

The cumulative **Risk Score** determines the final instruction given to the user:

Risk Level	Risk Score	Final Verdict
High	60+	POTENTIAL SCAM
Medium	35 - 59	VERIFY CAREFULLY
Low	< 35	LIKELY LEGITIMATE

30

5. PRODUCT DESCRIPTION ANALYSIS



31

6. BROWSER EXTENSION

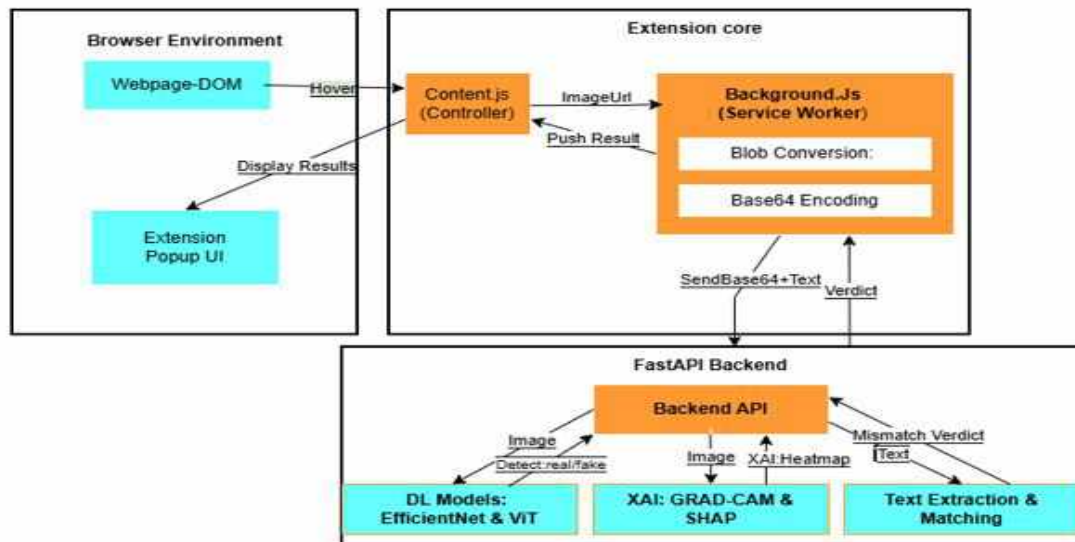
- Developed to enable real-time fake image detection on online marketplaces.
- Allows users to check image authenticity directly during browsing.
- Works without modifying the existing marketplace infrastructure.

Working Process

- Product images are extracted from the webpage
- Images are preprocessed within the extension
- Preprocessed images are sent to the backend via REST-based API
- EfficientNet and ViT models perform inference
- System returns:
 - Classification result (Real / Fake)
 - Confidence score
 - Explainable AI (XAI) results

32

6. BROWSER EXTENSION



33

Results

34

CNN

Dataset Information:

Training samples: 13456

Validation samples: 3214

Test samples: 3217

Classes: Fake, Real

Confusion Matrix:

[1490 10]

[11 1706]

Training vs Validation Performance

Best Validation Accuracy: 99.25%

Best Validation Loss: 0.0315

Total Training Time: 55.7 minutes

Class	Precision	Recall	F1-Score	Support
Fake	0.9927	0.9933	0.9930	1500
Real	0.9942	0.9936	0.9939	1717

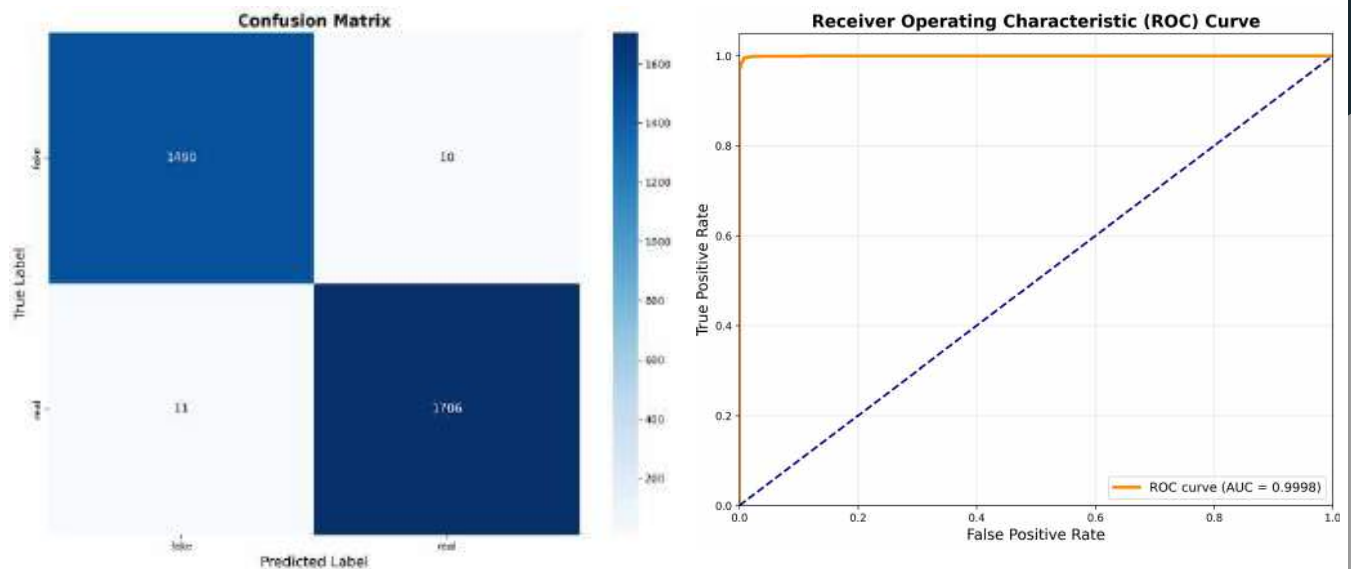
Test Results:

Test Accuracy: 99.3472%

ROC AUC Score: 0.9998

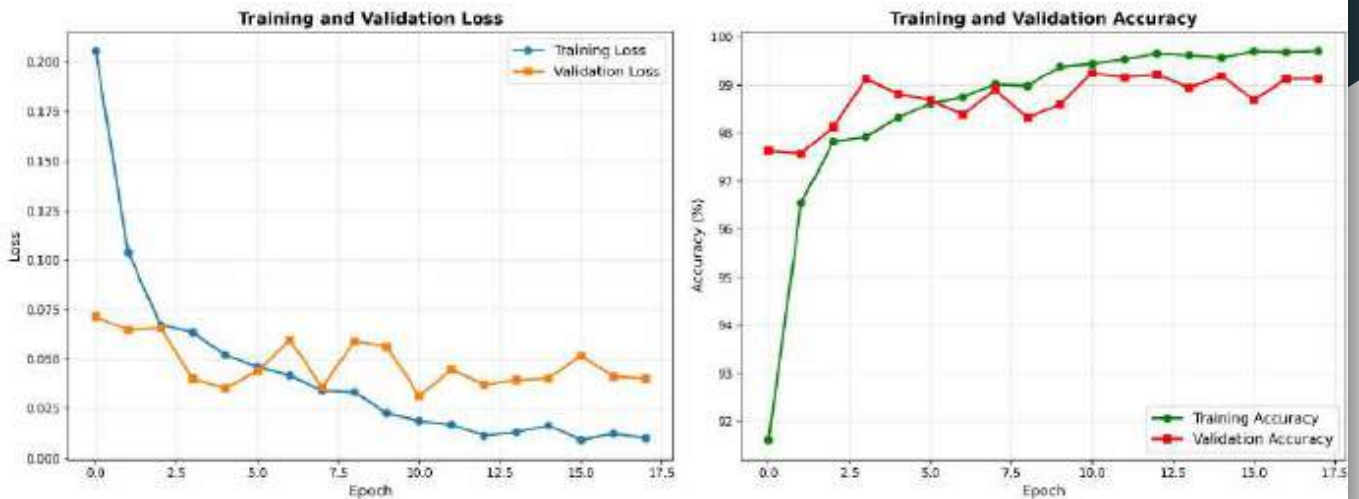
35

CNN



36

CNN



37

ViT

Dataset Information:

Training samples: 13456
 Validation samples: 3214
 Test samples: 3217
 Classes: Fake, Real

Confusion Matrix:

[1478 22]
 [10 1707]

Training vs Validation Performance

Best Validation Accuracy: 98.66%
 Best Validation Loss: 0.0694
 Total Training Time: 238.8 minutes

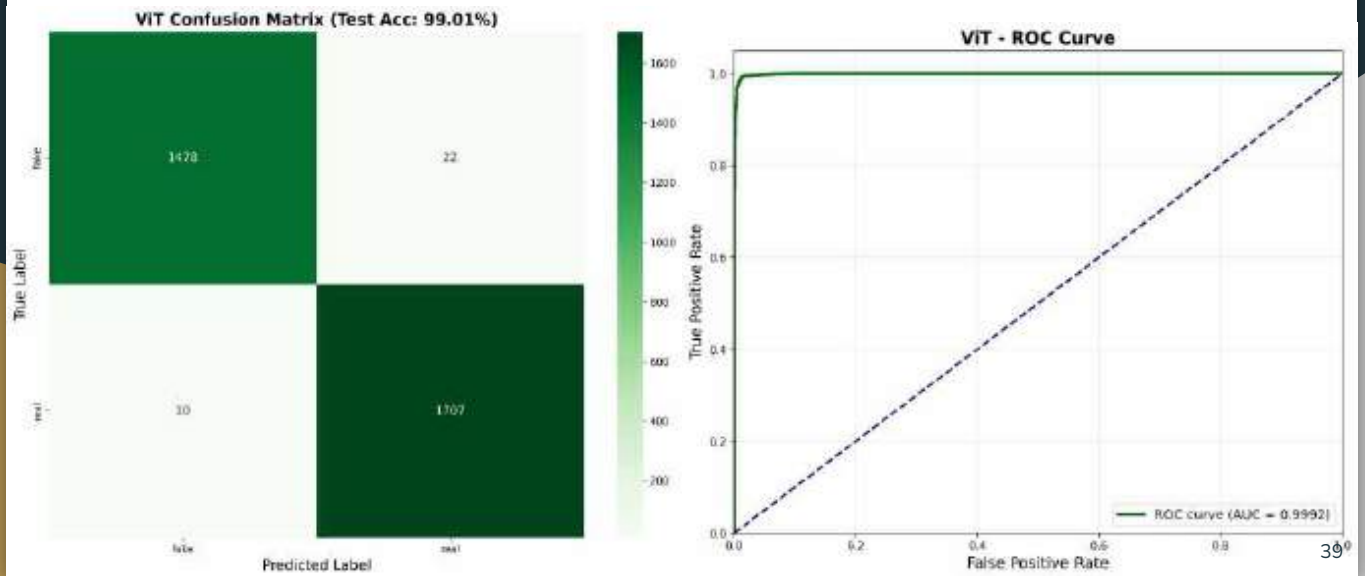
Class	Precision	Recall	F1-Score	Support
Fake	0.9933	0.9853	0.9893	1500
Real	0.9873	0.9942	0.9907	1717

Test Results:

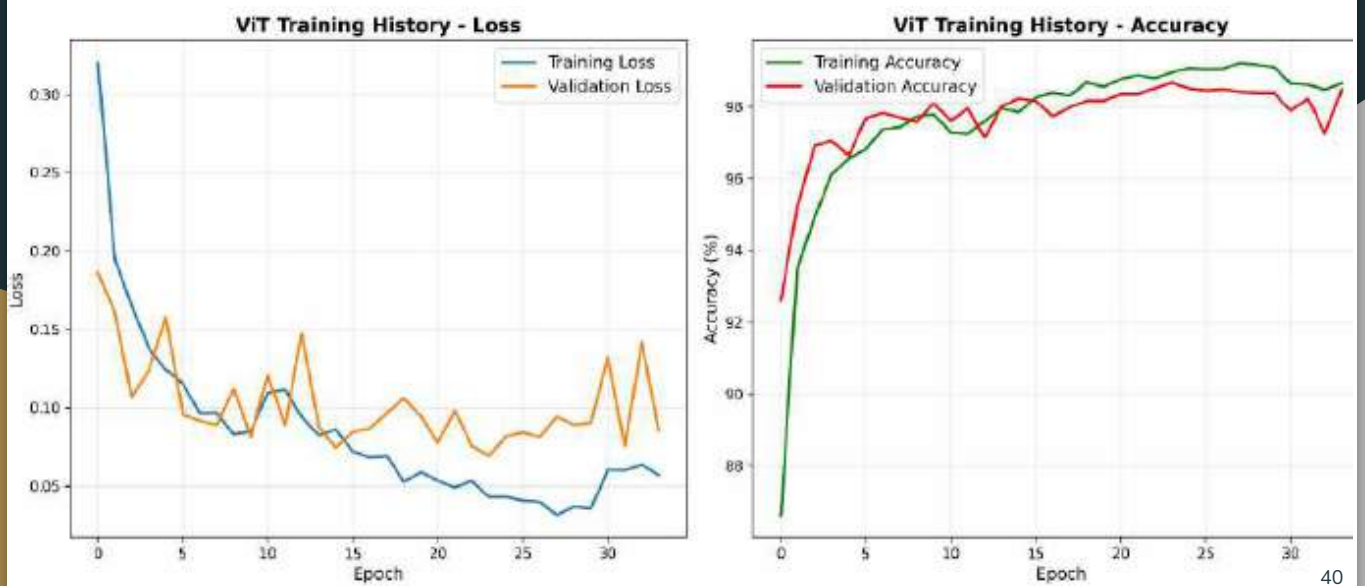
Test Accuracy: 99.0053%
 ROC AUC Score: 0.9992

38

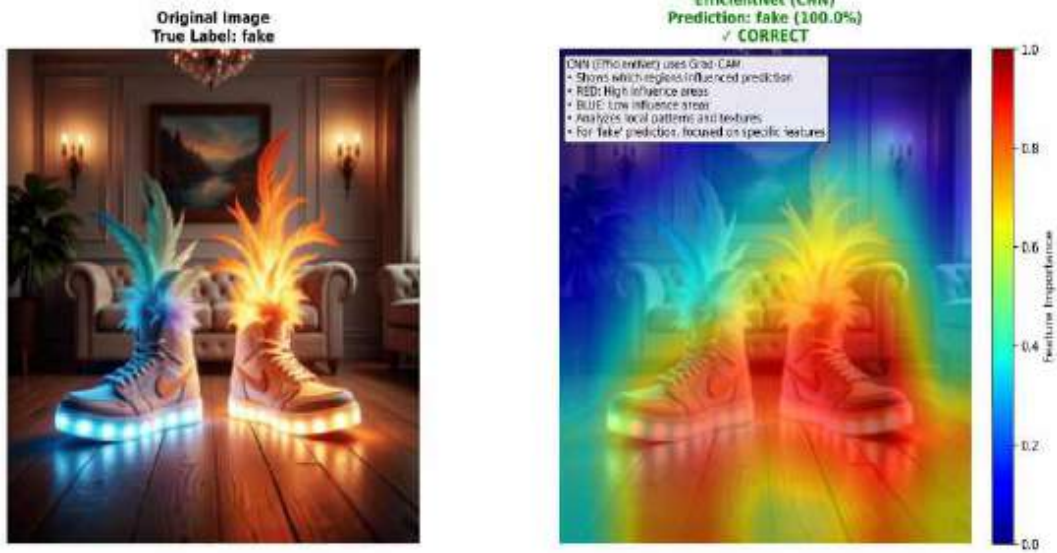
ViT



ViT

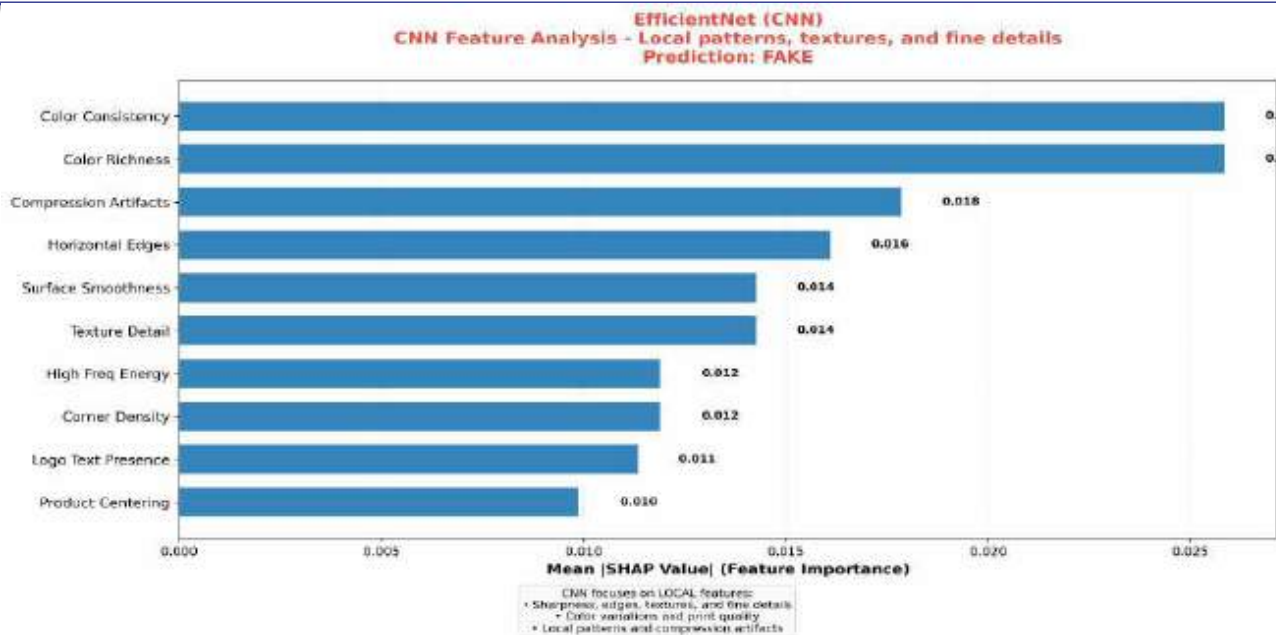


GRAD-CAM



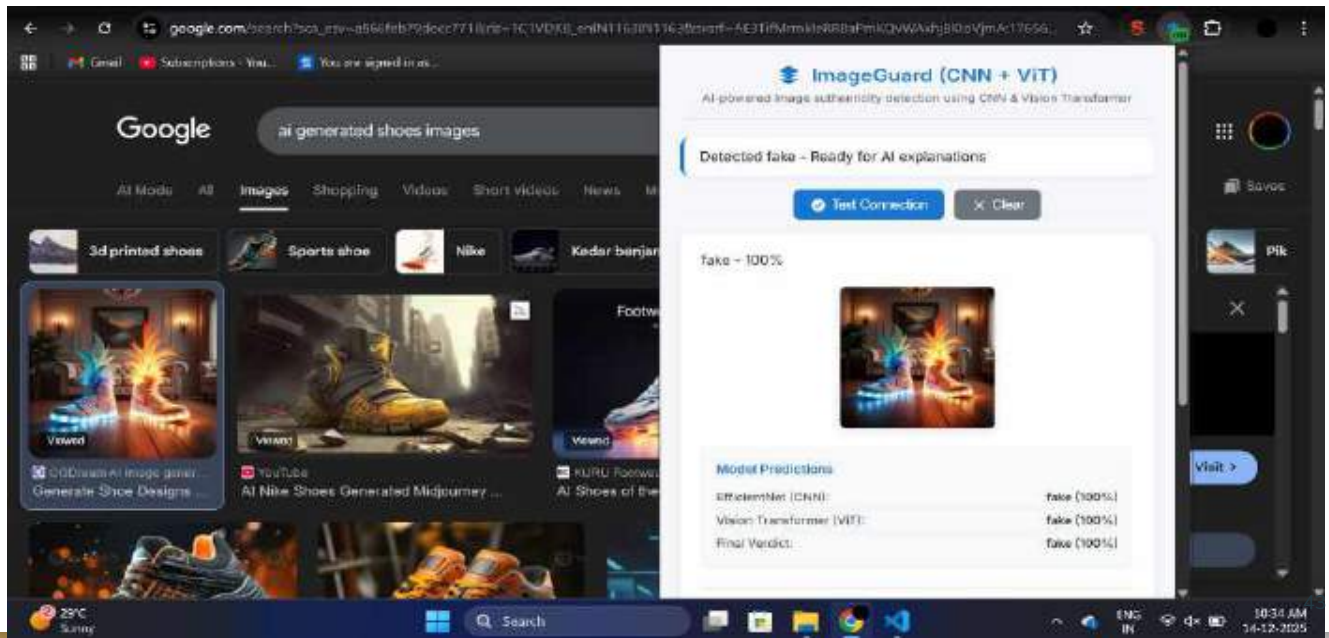
41

SHAP



42

BROWSER EXTENSION



RESULTS-CNN (Real)

```
(venv_pytorch) PS D:\project> & D:/project/venv_pytorch/Scripts/Activate.ps1
>> python D:\project\predict_new_image.py
Using device: cuda

Loading EfficientNet-B2 model...
✓ EfficientNet-B2 loaded successfully
Model accuracy: 99.72677595628416
Classes: ['fake', 'real']

=====
EFFICIENTNET-B2 IMAGE PREDICTION SYSTEM
=====
Model: EfficientNet-B2 (99.74% test accuracy)
Device: cuda
=====

MAIN MENU
=====
1. Predict single image
2. Batch predict from folder
3. Show prediction statistics
4. Reset statistics
5. Test on sample images
6. Exit
=====

Select option (1-6): 1
Enter image path: C:\Users\ASUS\Downloads\image1.jpg
```

CNN test image
(Real):



=====

CNN PREDICTION RESULTS

=====

Image: image1.jpg

Prediction: REAL

Confidence: 100.00%

Detailed Probabilities:

Fake: 0.00%

Real: 100.00%

=====

RESULTS-CNN (Fake)

```
(venv_pytorch) PS D:\project> & D:/project/venv_pytorch/Scripts/Activate.ps1
>> python D:\project\predict_new_image.py
Using device: cuda

Loading EfficientNet-B2 model...
✓ EfficientNet-B2 loaded successfully
Model accuracy: 99.72677595628416
Classes: ['fake', 'real']

=====
EFFICIENTNET-B2 IMAGE PREDICTION SYSTEM
=====
Model: EfficientNet-B2 (99.74% test accuracy)
Device: cuda

=====
MAIN MENU
=====
1. Predict single image
2. Batch predict from folder
3. Show prediction statistics
4. Reset statistics
5. Test on sample images
6. Exit
=====
Select option (1-6): 1
Enter image path: C:\Users\ASUS\Downloads\image.jpg
```

CNN test image
(Fake):



CNN PREDICTION RESULTS

Image: image.jpg

Prediction: FAKE

Confidence: 99.99%

Detailed Probabilities:

Fake: 99.99%

Real: 0.01%

45

RESULTS-ViT (Real)

```
(venv_pytorch) PS D:\project> & D:/project/venv_pytorch/Scripts/Activate.ps1
>> python D:\project\check_vit.py
Using device: cuda
GPU: NVIDIA GeForce RTX 3050 Laptop GPU

Loading Vision Transformer (ViT-Base) model...
✓ Loaded best ViT model
Best validation accuracy: 97.45%
✓ ViT model loaded and ready for inference
Device: cuda
Parameters: 88,327,298

=====
VISION TRANSFORMER (ViT) PREDICTION SYSTEM
=====
Model: ViT-Base Patch16 224
Device: cuda

=====
ViT MAIN MENU
=====
1. Predict single image
2. Batch predict from folder
3. Show prediction statistics
4. Reset statistics
5. Test on sample images
6. Exit
=====
Select option (1-6): 1
Enter image path: C:\Users\ASUS\Downloads\image1.jpg
```

ViT test image
(Real):



ViT PREDICTION RESULTS

Image: image1.jpg

Prediction: real

Confidence: 99.92%

Fake Probability: 0.08%

Real Probability: 99.92%

Processing Time: 299.2 ms

46

RESULTS-ViT (Fake)

```
(venv_pytorch) PS D:\project> & D:/project/venv_pytorch/Scripts/Activate.ps1
>> python D:\project\check_vit.py
Using device: cuda
GPU: NVIDIA GeForce RTX 3050 Laptop GPU

Loading Vision Transformer (ViT-Base) model...
✓ Loaded best ViT model
  Best validation accuracy: 97.45%
✓ ViT model loaded and ready for inference
  Device: cuda
  Parameters: 86,327,298

=====
VISION TRANSFORMER (ViT) PREDICTION SYSTEM
=====
Model: ViT-Base Patch16 324
Device: cuda
=====

=====
ViT MAIN MENU
=====
1. Predict single image
2. Batch predict from folder
3. Show prediction statistics
4. Reset statistics
5. Test on sample images
6. Exit
=====
Select option (1-6): 1
Enter image path: C:\Users\ASUS\Downloads\image.jpg
```

ViT test image
(Fake):



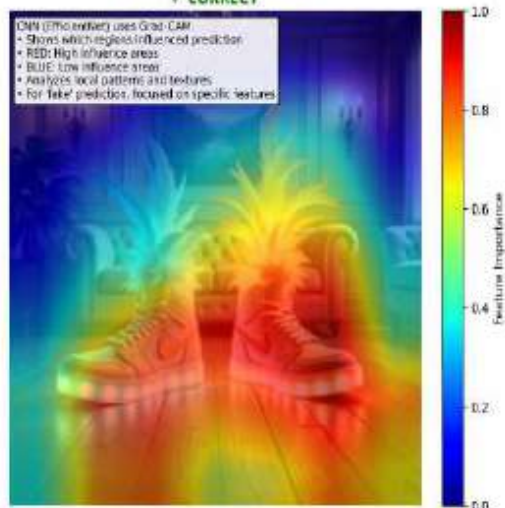
```
=====
ViT PREDICTION RESULTS
=====
Image: image.jpg
Prediction: fake
Confidence: 99.96%
Fake Probability: 99.96%
Real Probability: 0.04%
Processing Time: 302.2 ms
=====
```

RESULTS-XAI (GRAD-CAM+CNN)

Original Image
True Label: fake

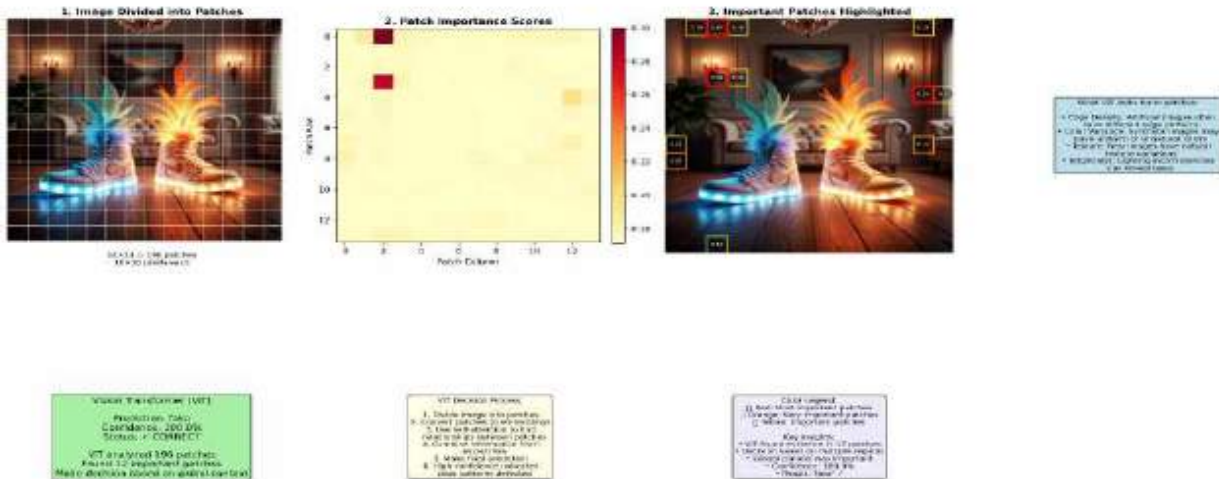


EfficientNet (CNN)
Prediction: fake (100.0%)
✓ CORRECT



RESULTS-XAI (GRADCAM+ViT)

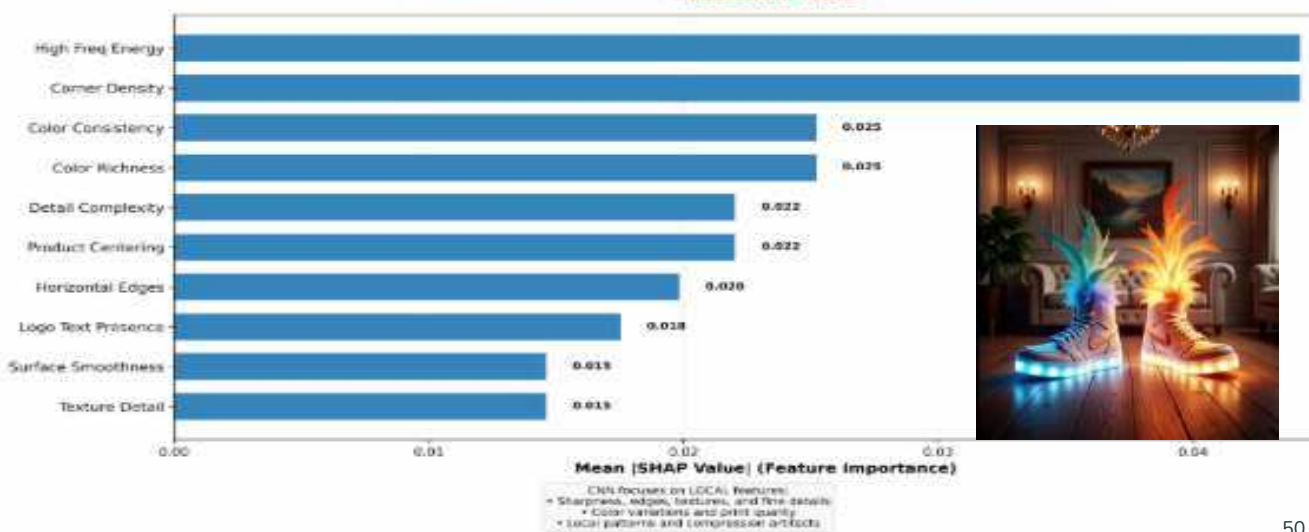
Vision Transformer (ViT) Deep Analysis: Why it predicted "Fake"



49

RESULTS-XAI (SHAP+CNN)

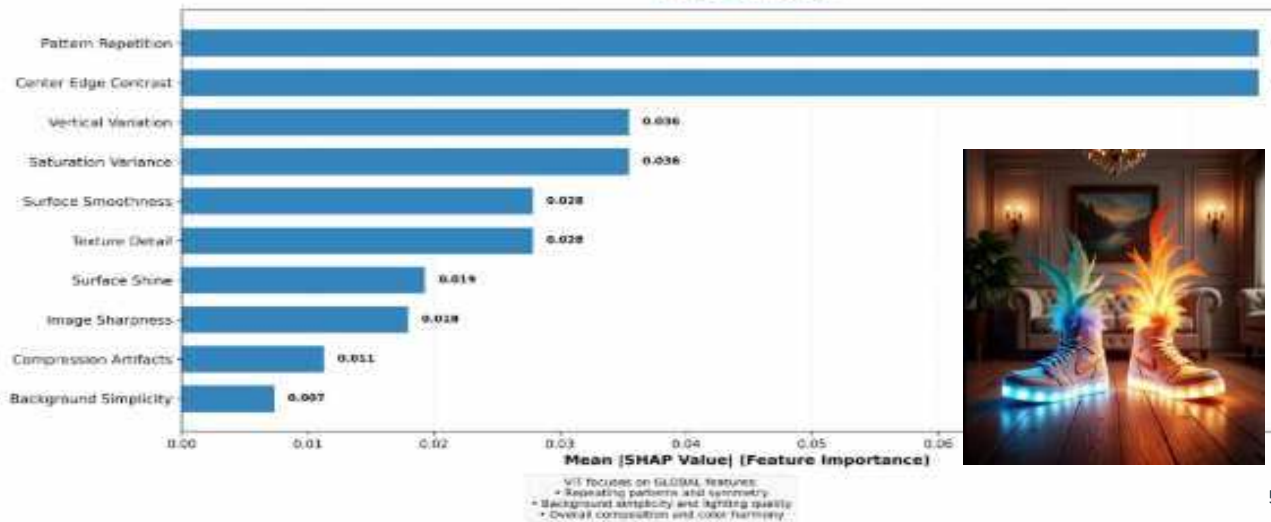
EfficientNet (CNN)
CNN Feature Analysis - Local patterns, textures, and fine details
Prediction: FAKE



50

RESULTS-XAI (SHAP+ViT)

Vision Transformer (ViT)
ViT Feature Analysis - Global composition, symmetry, and relationships
Prediction: FAKE



51

RESULTS-BROWSER EXTENSION (Real)

amazon.in/nike-shoes/5?k=nike+shoes

Up to ₹4,300
 ₹4,300 - ₹5,300
 ₹5,300 - ₹6,500
 ₹6,500 - ₹9,000
 Over ₹9,000

Deals & Discounts
 All Discounts
 Today's Deals

Brands
☐ Nike
☐ Campus
☐ Boldfit
☐ new balance
☐ Bacco Bucci
☐ Reebok
☐ DOCTOR EXTRA SOFT
 See more

Women's Shoe Size (UK/India)
 2.5 3 3.5 4 4.5

Results
 Check each product page for other buying options. Price and availability may vary.

Previously viewed

Sponsored @
Boldfit
 BoldFit Running Shoes for Men
 Soft Sole Sports Shoes for Me...
 4.0 ★★★★★ (540)
 200+ bought in last month

Sponsored @
CULT
 Men's Zephyr R...
 3.7 ★★★★★ (41)
 ₹2,249 (M.R.P.)

ImageGuard (CNN + ViT)
 AI-powered image authenticity detector using CNN & Vision Transformer

Detected real - Ready for AI explanations

Test Connection X Clear

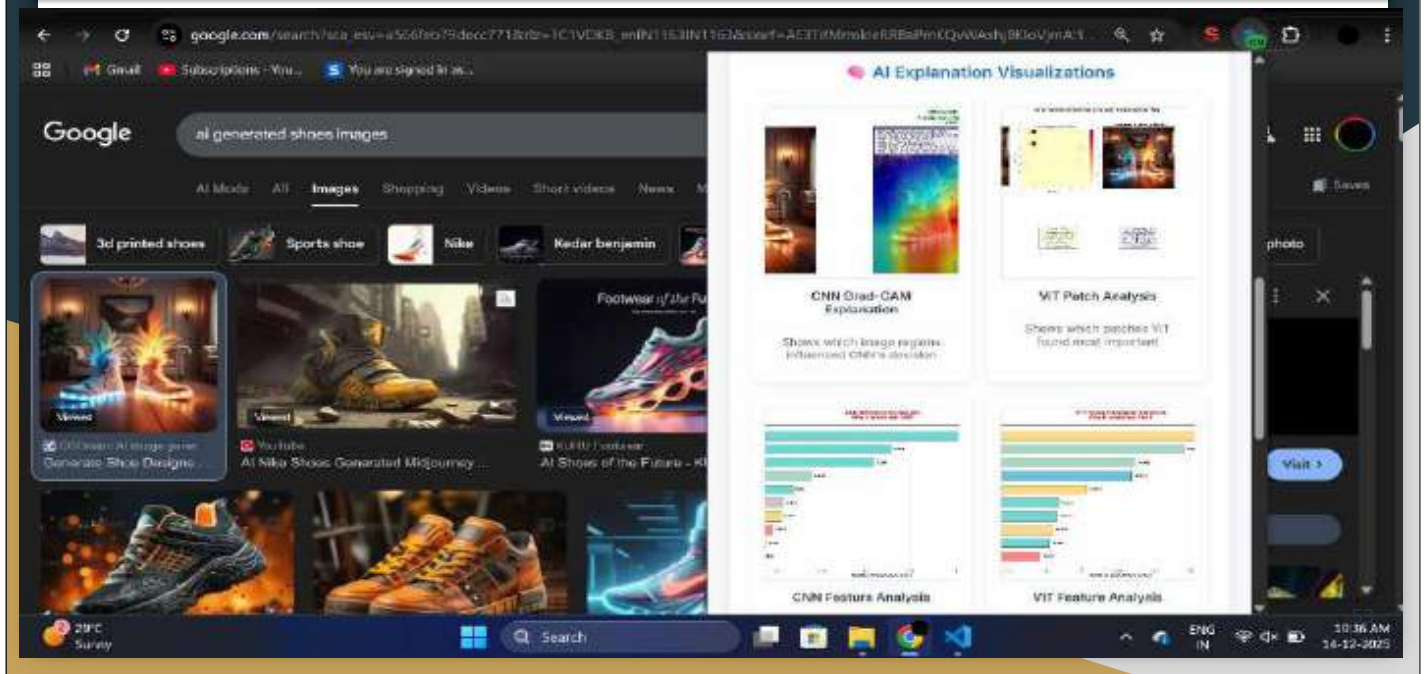
real - 100%

Model Predictions

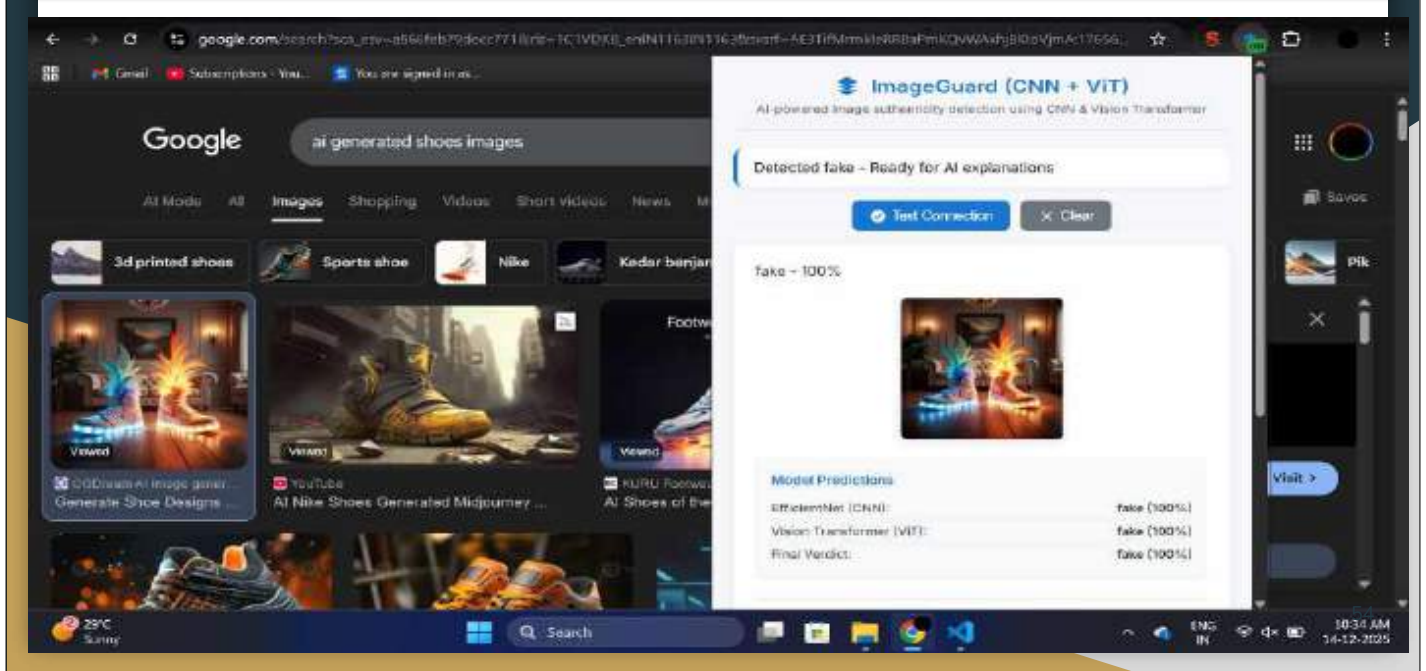
EfficientNet (CNN)	real (100%)
Vision Transformer (ViT)	real (100%)
Final Verdict:	real (100%)

52

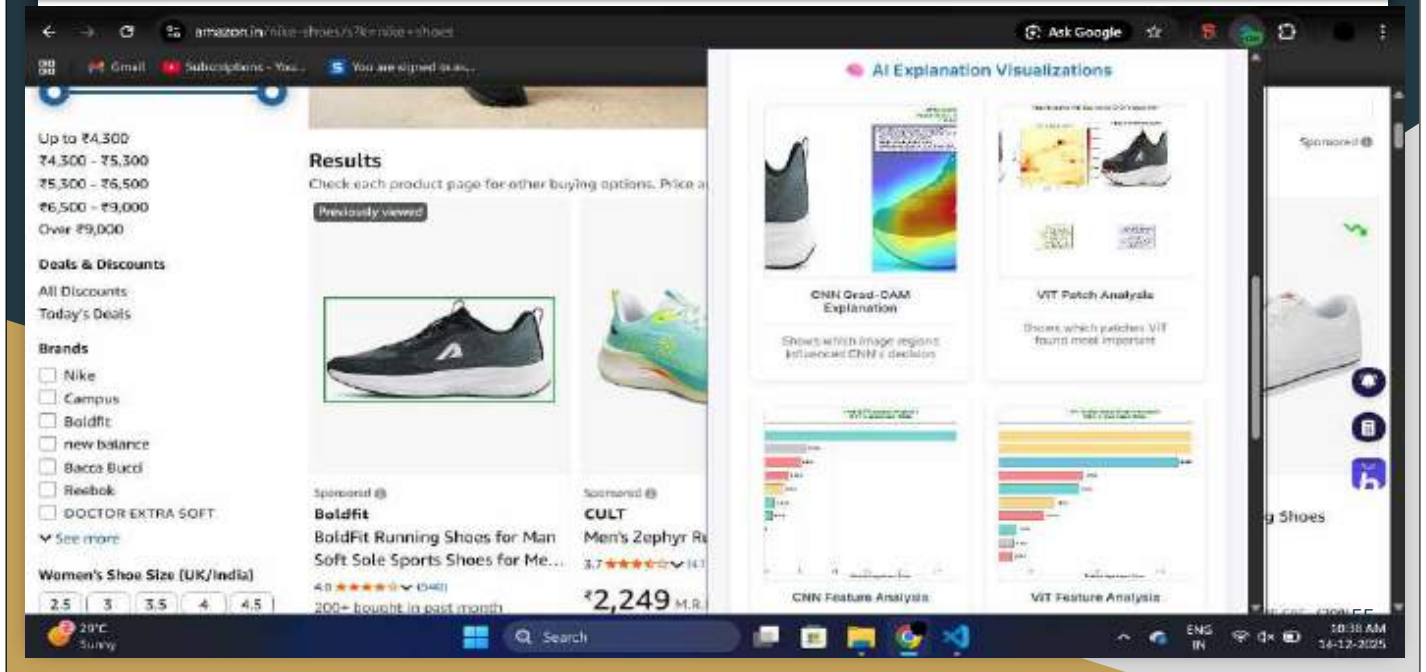
RESULTS-BROWSER EXTENSION (Real)



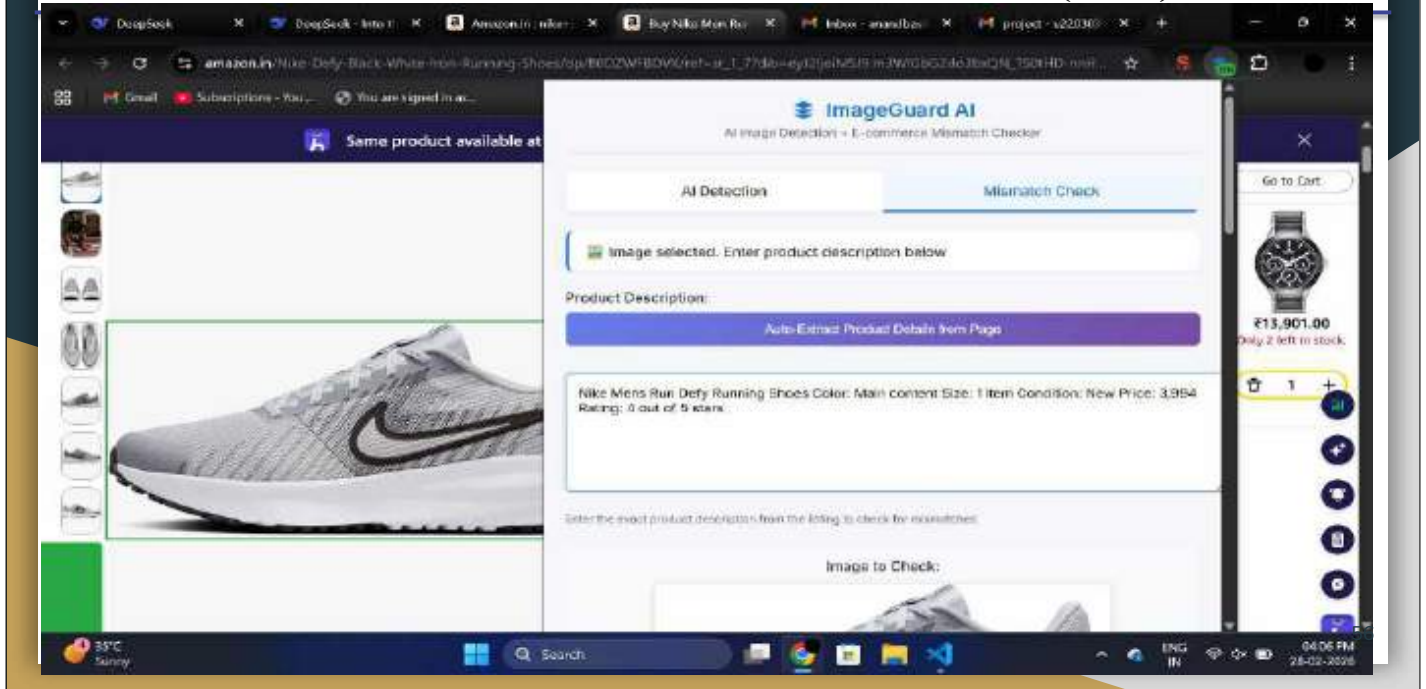
RESULTS-BROWSER EXTENSION (Fake)



RESULTS-BROWSER EXTENSION (Fake)



PRODUCT DESCRIPTION ANALYSIS (Real)



PRODUCT DESCRIPTION ANALYSIS (Real)

The screenshot shows an Amazon India product page for a Nike sneaker. A central overlay titled "E-commerce Mismatch Detection" displays a "LOW RISK - LIKELY LEGITIMATE" status with a score of 1.00, labeled as an "Excellent Match". The overlay explains that the score ranges from 0 (completely different) to 1 (perfect match). Below the score, it provides a breakdown of the match analysis:

Category	Match Status
Image Shows	% Shoe/Footwear
Description Type	% Shoe/Footwear
Image Colors	White (66.2%), Gray (26.2%), Dark Gray (7.8%)
Description Color	Not specified

The background shows the product image of a gray and white Nike sneaker, a sidebar with other shoe options, and a shopping cart on the right containing a watch for ₹13,901.00.

PRODUCT DESCRIPTION ANALYSIS (Real)

This screenshot shows the same Amazon India product page with a more detailed analysis overlay. The overlay is divided into several sections:

- Color Analysis:**
 - White: 66.2%
 - Gray: 26.2%
 - Dark Gray: 7.8%
 - Main colors: White (66.2%), Gray (26.2%), and Dark Gray (7.8%)
- Quality Analysis:**
 - Overall Quality: Good
 - Image Score: 2400
 - Image Level: Very High
 - Star Level: Very Low
 - Confidence: 62
 - Resolution: 224 x 224
 - Note: Good quality image suitable for e-commerce
- Background:**
 - Background Type: Complex/Busy
 - Product Photo: Yes ✓
 - Confidence Score: 94.4%
 - Simplicity: 45.8%
 - White Background: 64%
 - Professional product photo: (checkbox)
- Details:**
 - Product Type: % Shoe/Footwear
 - Brand: Nike
 - Condition: New
 - Size: 1
 - Price: 3894
 - Link: Nike - Shoe - Size 1

The background of the page remains the same, showing the sneaker product image and the shopping cart with the watch.

PRODUCT DESCRIPTION ANALYSIS (Fake)

The screenshot shows a Google search for "ai generated shoes". The search results include a video thumbnail of a yellow and black sneaker, a YouTube video titled "AI Nike Shoes Generated Midjourney...", and a thumbnail of a purple and orange sneaker. A product analysis overlay is displayed on the right side of the screen.

MEDIUM RISK - VERIFY CAREFULLY

45

0.20 ⚠️ ⚠️ **Poor Match**
The product shows significant differences from the description.

How we calculate match: We compare product type and color from the image with what's described. The score ranges from 0 (completely different) to 1 (perfect match). Higher scores mean the product matches what's described.

Image Shows	Description Says
Shoe/Footwear	Watch/Timepiece

Image Colors	Description Color
Navy Blue (35.4%), Navy Blue (33.4%), Sky Blue (19.4%), Sage Green (11.7%)	Not specified

Product Match: ⚠️ ⚠️ **Poor Match**

PRODUCT DESCRIPTION ANALYSIS (Fake)

The screenshot shows a Google search for "ai generated shoes". The search results include a video thumbnail of a yellow and black sneaker, a YouTube video titled "AI Nike Shoes Generated Midjourney...", and a thumbnail of a purple and orange sneaker. A detailed product analysis overlay is displayed on the right side of the screen.

Product Match: ⚠️ ⚠️ **Poor Match**
The product shows significant differences from the description. Match Score: 0.20 (0-1 scale, higher is better)

- 1. **Product Type Mismatch:** Image shows: Shoe/Footwear but description mentions: Watch/Timepiece
- 2. **Product type mismatch:** Image shows: Shoe/Footwear but description mentions: watch
- 3. **Product type mismatch:** Image shows shoe but description mentions: watch

⚠️ No price mentioned in description

Color Analysis	
Navy Blue	35.4%
Navy Blue	33.4%
Sky Blue	19.4%
Sage Green	11.7%

Main colors: Navy Blue (35.4%), Navy Blue (33.4%) and 3 other shades

Quality Analysis	
Overall Quality: Good	Sharpness: 947
Blur Level: Very High	Distortion: Very Low
Graininess: 56	Resolution: 224x224

Good quality image suitable for e-commerce

ASSUMPTIONS

- **Model Accuracy:** The Deep Learning models (CNNs and Vision Transformers) will effectively detect fake, edited, or AI-generated images.
- **Reliable Explanations:** The Explainable AI tools (Grad-CAM and SHAP) will correctly highlight the suspicious parts of an image.
- **Real-Time Performance:** The system will process and display results quickly, without affecting the user's browsing experience.
- **User Adoption:** Users will install the extension and may contribute by reporting incorrect results.
- **Cross-Platform Support:** The browser extension will work well across various e-commerce websites and browsers.

61

RISK AND CHALLENGES

- **Evolving Threats:** As generative AI advances, new types of fake images may become harder to detect. Attackers might also attempt adversarial attacks to bypass the system.
- **Data & Accuracy Limitations:** The training dataset may not cover all manipulation types, potentially causing false positives (flagging real images) or false negatives (missing fakes).
- **Performance & Maintenance:** Real-time detection could affect browsing speed, and the system will require regular updates to maintain accuracy and platform compatibility.
- **Privacy & Cost Concerns:** Processing images from users' browsers raises privacy issues, and GPU requirements may increase the cost of large-scale deployment.

62

CONCLUSION

- ImageGuard combines Deep Learning and Explainable AI (XAI), to effectively detect fake or misleading product images in e-commerce platforms.
- A deep learning model capable of detecting fake and manipulated images in online marketplaces.
- Integration of Explainable AI (XAI) ensured that users could understand and trust the system's decisions.
- Implemented a browser extension for real-time, user-friendly deployment
- The project contributes to enhancing the authenticity and trustworthiness of online marketplaces by providing a reliable tool for fraud detection.

63

FUTURE PLANS

- Extend the dataset to include additional product categories.
- The system can be extended to support cross-platform deployment, including mobile applications, to increase accessibility for end users.
- Continuous learning mechanisms can be introduced using user feedback to improve detection accuracy over time.
- The model can be adapted to detect emerging AI-generated image techniques by periodic retraining with newly collected data.
- Improve system performance and scalability through further model tuning and larger datasets.

64

PUBLICATIONS

ImageGuard: An Explainable Hybrid Deep Vision Framework for Counterfeit Image Detection in E-Commerce Marketplaces : Communicated to Ceylon Journal of Science-Scopus Indexed

65

REFERENCES

- [1] AlShariah, N.M., Khader, A. and Saudagar, J., 2019. Detecting fake images on social media using machine learning. *International Journal of Advanced Computer Science and Applications*, 10(12), pp.170-176.
- [2] Ghita, B., Kuzminykh, I., Usama, A., Bakhshi, T. and Marchang, J., 2024, June. Deepfake image detection using vision transformer models. In *2024 IEEE International Black Sea Conference on Communications and Networking (BlackSeaCom)* (pp. 332-335). IEEE.
- [3] Van Zyl, C., Ye, X. and Naidoo, R., 2024. Harnessing eXplainable artificial intelligence for feature selection in time series energy forecasting: A comparative analysis of Grad-CAM and SHAP. *Applied Energy*, 353, p.122079.
- [4] Štancel, M. and Hulič, M., 2019. An introduction to image classification and object detection using YOLO detector. In *CEUR workshop proceedings* (Vol. 2403, pp. 1-8).
- [5] Iqbal, J.M., Manikandan, T., Tamilselvan, K.S., Muruganandham, A. and Karthikeyan, S., 2024, July. BERT-Based Deep Learning Models for Enhanced Object Detection in Images. In *2024 Second International Conference on Advances in Information Technology (ICAIT)* (Vol. 1, pp. 1-6). IEEE.

66

THANK YOU

Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

- 1. Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
- 2. Problem analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

3. Design/development of solutions:Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

4. Conduct investigations of complex problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and; interpretation of data to provide valid conclusions.

5. Engineering Tool Usage:Create, select and apply appropriate techniques, resources and modern engineering and; IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

6. The Engineer and the world: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

7. Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and; international laws.

8. Individual and Collaborative Team Work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

9. Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

10. Project management and finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

11. Life-long learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

Programme Specific Outcomes (PSO)

A graduate of the Computer Science and Engineering Program will demonstrate:

PSO1: Computer Science Specific Skills

The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.

PSO2: Programming and Software Development Skills

The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry.

PSO3: Professional Skills

The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur.

Course Outcomes (CO)

After the completion of the course the student will be able to:

CO1: Model and solve real world problems by applying knowledge across domains (Cognitive knowledge level: Apply).

CO2: Develop products, processes or technologies for sustainable and socially relevant applications (Cognitive knowledge level: Apply).

CO3: Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks (Cognitive knowledge level: Apply).

CO4: Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms (Cognitive knowledge level: Apply).

CO5: Identify technology/research gaps and propose innovative/creative solutions (Cognitive knowledge level: Analyze).

CO6: Organize and communicate technical and scientific findings effectively in written and oral forms (Cognitive knowledge level: Apply).

Appendix C: CO-PO-PSO Mapping

COURSE OUTCOMES:

SNO	DESCRIPTION	BLOOM'S TAXONOMY LEVEL
1	Model and solve real world problems by applying knowledge across domains.	Apply
2	Develop products, processes or technologies for sustainable and socially relevant application.	Apply
3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks.	Apply
4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms.	Apply
5	Identify technology/research gaps and propose innovative/creative solutions.	Analyze
6	Organize and communicate technical and scientific findings effectively in written and oral forms.	Apply

CO-PO AND CO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	2	1		2	1	1	1	1	1	2	2	2
CO2	2	2	2		2	3	2	1	1	1	1	2	2	
CO3								3	2	2	1			3
CO4		1	2		3	2	3	2	2	3	1		2	2
CO5	2	3	3	1	2					1	3	3		
CO6			2			2	2	2	3	1	1			3

Table 1: CO-PO-PSO Mapping

JUSTIFICATIONS FOR CO-PO MAPPING

Mapping	LOW/ MEDIUM /HIGH	Justification
101003/ CS822U.CO1- PO1	M	Applies engineering knowledge, mathematics, and computing fundamentals to model and solve real-world problems.
101003/ CS822U.CO1- PO2	M	Requires identification, formulation and analysis of engineering problems before developing solutions.

Mapping	LOW/ MEDIUM /HIGH	Justification
101003/ CS822U.CO1- PO3	M	Involves designing systems or solutions to address identified engineering needs.
101003/ CS822U.CO1- PO4	L	Limited investigation or experimentation may be required to validate the developed models or solutions.
101003/ CS822U.CO1- PO6	M	Solutions developed may influence societal, environmental, or sustainability aspects of engineering applications.
101003/ CS822U.CO1- PO7	M	Encourages consideration of ethical and professional responsibilities while developing solutions.
101003/ CS822U.CO1- PO8	L	Some level of teamwork may be required while working on modeling or project tasks.
101003/ CS822U.CO1- PO9	L	Requires communication of results and findings through reports or presentations.
101003/ CS822U.CO1- PO10	L	Involves basic planning and execution of tasks while implementing solutions.
101003/ CS822U.CO1- PO11	L	Encourages independent learning and adapting new knowledge to solve real-world problems.
101003/ CS822U.CO1- PSO1	M	Uses core computer science principles to analyze problems and design appropriate technical solutions.
101003/ CS822U.CO1- PSO2	M	Involves algorithm design and application of software development practices to implement solutions.
101003/ CS822U.CO1- PSO3	M	Applies computer science fundamentals in research or innovative problem solving for societal needs.
101003/ CS822U.CO2- PO1	M	Utilizes engineering knowledge to design and develop products or systems.

Mapping	LOW/ MEDIUM /HIGH	Justification
101003/ CS822U.CO2- PO2	M	Requires analysis of user needs and technological feasibility before development.
101003/ CS822U.CO2- PO3	M	Focuses on design and development of solutions addressing real-world requirements.
101003/ CS822U.CO2- PO5	M	Requires use of modern engineering and IT tools for system or product development.
101003/ CS822U.CO2- PO6	H	Emphasizes sustainability and societal impact during solution development.
101003/ CS822U.CO2- PO7	M	Ethical and professional responsibilities are considered when developing socially relevant technologies.
101003/ CS822U.CO2- PO8	L	Development activities may require collaboration among team members.
101003/ CS822U.CO2- PO9	L	Involves communicating design ideas and development outcomes.
101003/ CS822U.CO2- PO10	L	Requires basic planning and coordination during development tasks.
101003/ CS822U.CO2- PO11	L	Encourages continuous learning of emerging technologies while building solutions.
101003/ CS822U.CO2- PSO1	M	Applies computer science fundamentals to design technically feasible solutions.
101003/ CS822U.CO2- PSO2	M	Uses programming and software development practices to implement solutions.
101003/ CS822U.CO3- PO8	H	Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.

Mapping	LOW/ MEDIUM /HIGH	Justification
101003/ CS822U.CO3- PO9	M	Requires effective communication among team members during collaboration.
101003/ CS822U.CO3- PO10	M	Team leadership involves project coordination and task management.
101003/ CS822U.CO3- PO11	L	Team interaction supports learning from peers and adapting new approaches.
101003/ CS822U.CO3- PSO3	H	Professional collaboration and teamwork support innovative computer science solutions.
101003/ CS822U.CO4- PO2	L	Requires analysis and planning before executing project tasks.
101003/ CS822U.CO4- PO3	M	Involves designing and implementing project solutions based on planned activities.
101003/ CS822U.CO4- PO5	H	Requires effective use of modern tools and resources during project execution.
101003/ CS822U.CO4- PO6	M	Considers societal and environmental impact while implementing solutions.
101003/ CS822U.CO4- PO7	H	Emphasizes adherence to ethical practices and professional responsibilities.
101003/ CS822U.CO4- PO8	M	Task execution may involve teamwork and coordination among team members.
101003/ CS822U.CO4- PO9	M	Requires communication of progress, outcomes and documentation.
101003/ CS822U.CO4- PO10	H	Focuses on project management including scheduling, resource allocation and task execution.

Mapping	LOW/ MEDIUM /HIGH	Justification
101003/ CS822U.CO4- PO11	L	Encourages adaptability and continuous learning during project implementation.
101003/ CS822U.CO4- PSO2	M	Uses software development practices to execute and manage project tasks.
101003/ CS822U.CO4- PSO3	M	Applies computer science knowledge in practical project development environments.
101003/ CS822U.CO5- PO1	M	Uses engineering knowledge to understand technological limitations and opportunities.
101003/ CS822U.CO5- PO2	H	Requires deep problem analysis and identification of re-search gaps.
101003/ CS822U.CO5- PO3	H	Proposes innovative design solutions to address identified gaps.
101003/ CS822U.CO5- PO4	L	May involve limited investigation or validation of proposed ideas.
101003/ CS822U.CO5- PO5	M	Uses appropriate tools and techniques to analyze and develop innovative solutions.
101003/ CS822U.CO5- PO10	L	Requires minimal planning while proposing new solution approaches.
101003/ CS822U.CO5- PO11	H	Strongly promotes independent learning and exploration of emerging technologies.
101003/ CS822U.CO5- PSO1	H	Applies computer science fundamentals to identify and solve complex technical problems.
101003/ CS822U.CO6- PO3	M	Requires presentation of design concepts and developed solutions.

Mapping	LOW/ MEDIUM /HIGH	Justification
101003/ CS822U.CO6- PO6	M	Communication may include discussion of societal or environmental implications.
101003/ CS822U.CO6- PO7	M	Requires responsible and ethical communication of technical results.
101003/ CS822U.CO6- PO8	M	Communication supports collaboration within team environments.
101003/ CS822U.CO6- PO9	H	Strongly relates to effective written and oral communication of technical findings.
101003/ CS822U.CO6- PO10	L	Requires structured reporting and documentation in project management contexts.
101003/ CS822U.CO6- PO11	L	Encourages continuous improvement in communication and presentation skills.
101003/ CS822U.CO6- PSO3	H	Communication of innovative computer science solutions and research outcomes.